MIRA CONTROL — functional hierarchy

22 consecutive circuit leaves: 100, 200, 300, 400 … 2200. Hierarchy pages are navigation, not additional circuitry.

MCU

Host processor, memory and supervision

STM32 supply, clocks and debug; FMC SDRAM; QSPI, microSD and host control. The same physical MCU spans three leaves.

INTERCONNECT

Power inputs and supervisory boundary

Explicit Control-side power and permission connector. Separate SYS3V3 and IO3V3 receivers prevent joining disabled receiver domains.

FPGA

Programmable logic, memory and external links

Artix-7 supplies, configuration, USB FIFO, DDR3L, Ethernet PHY and buffered MCU link. All FPGA sections belong to one physical package.

DAC

Analog commands and source buffers

Thirty-two BL voltages plus service voltage and current commands. The shared source-side CV/CC regulator is on sheet 2100; polarity is selected by matrix routing.

MATRIX

Held routing, line drivers and crossbar socket

Captured mode and address fields, 80-bit image, service bank, polarity and READ/WRITE return paths. Any active request mismatch inhibits RUN and raises configuration trip.

ADC

Current conversion and precision acquisition

TIA ranges and monitor selection, high-impedance compliance buffers, balanced differential ADC drive, reference and converter.

PROTECTION

Shared CV/CC source and hardware fault memory

Source-side Kelvin current sensing, shared PI control with bounded command and reset, independent backup overcurrent trip, configuration fault latch and external inhibit.

Physical references use owner-sheet base … base+99. Shared multi-unit packages retain one reference across their leaves.

Electrical repairs and an explicit change log are included. Read docs/REPAIR_NOTES_RU.md and the updated separate PDF before PCB release.

MIRA

Sheet: /
File: MIRA_Control.kicad_sch

Title: MIRA Control functional hierarchy

Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 1/30

MCU — Host processor, memory and supervision

Consecutive circuit order; select a child sheet to inspect its circuit and sheet-specific explanation.

100_MCU_POWER_CLOCKS_AND_DEBUG

STM32 supply, LDO reservoir, clocks, reset and SWD; DONE/INIT status inputs.

200_MCU_FMC_AND_SDRAM

MCU external SDRAM and FMC signals, defaults and local bypass.

300_MCU_QSPI_MICROSD_AND_CONTROL

MCU QSPI flash, microSD card detect and supervisory GPIO/UART links.

Group scope

STM32 supply, clocks and debug; FMC SDRAM; QSPI, microSD and host control. The same physical MCU spans three leaves.

Reading guide

Each numeric child contains its own circuit explanation. Identical global labels connect between leaves. A shared U reference denotes one physical multi-unit device.

Physical references follow the first circuit leaf containing the component; multi-unit devices are not duplicated.

MIRA

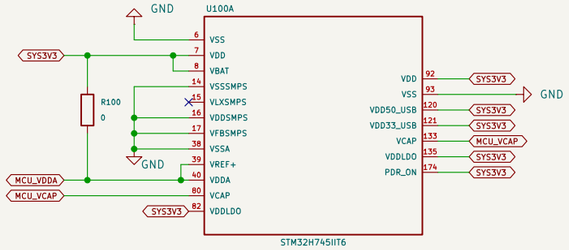
Sheet: /MCU/
File: MCU.kicad_sch

Title: Host processor, memory and supervision

Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 2/30

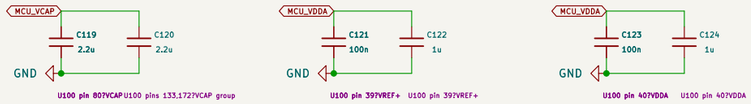
STM32 supply, clocks, reset and debug

Main MCU supply and core regulator | Rev.A

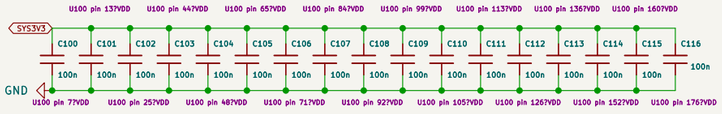


Main MCU supply bypass capacitors

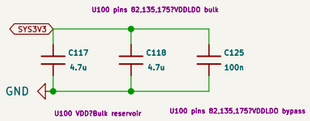
Core regulation and analog supply bypass



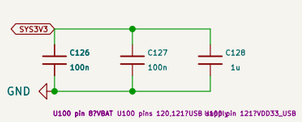
Main MCU supply bypass capacitors | Rev.A



VDD and LDO reservoir capacitors

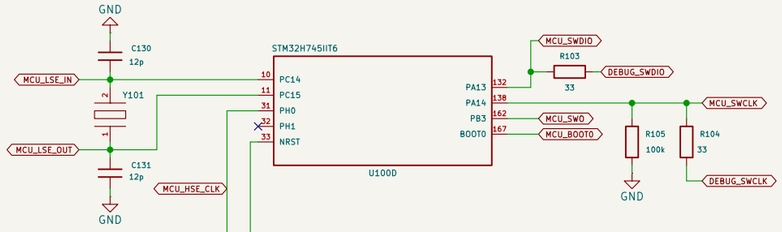


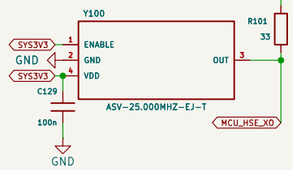
Backup and USB supply bypass



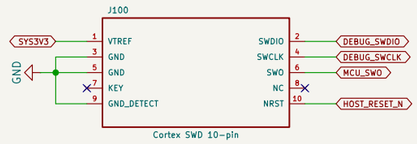
Main MCU clock references

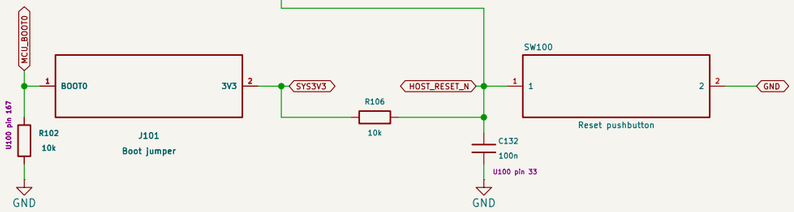
Main MCU clock references | Rev.A





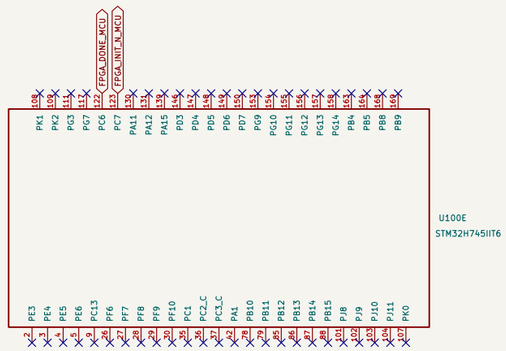
Main MCU reset, boot and SWD | Rev.A





MCU GPIO and FPGA configuration status

MCU GPIO and FPGA configuration status | Rev.A



Filtered supply: source declaration for ERC



Processor and supply domains

U100 is the STM32H745 host processor. SYS3V3 supplies VDD, VDDLDO, VBAT and the shown USB supply pins. R100 is a 0 ohm link feeding MCU_VDDA for VDDA and VREF+. The separate MCU units on the other sheets belong to this same physical package.

Internal core regulator

The three VCAP pins share MCU_VCAP, with C119 and C120 providing two 2.2 uF capacitors to ground. This net is the internal regulator reservoir, not an external supply output. VDDSMPS, VFBSMPS and VSSSMPS are grounded; VLXSMPS is left open in the drawn LDO supply configuration.

Local supply bypass

The 100 nF capacitors provide local bypass paths for the individual supply pins. The 4.7 uF capacitors support the shared rail. The analog, backup and USB supply groups have their own capacitors; their short return paths must be preserved during PCB placement.

Clock sources

Y100 provides a 25 MHz external clock through the 33 ohm R101 to MCU_HSE_CLK. Y101 is the 32.768 kHz LSE crystal, with 12 pF load capacitors C130 and C131. These two clock networks have different purposes and must be configured accordingly in firmware.

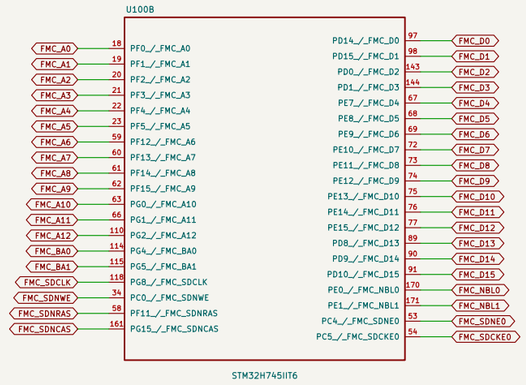
Reset, boot and access

R106 and C132 bias and filter HOST_RESET_N; SW100 pulls it low. R102 holds BOOT0 low, while J101 permits a high strap. J100 exposes SWD, SWO, reset and the supply reference. R103 and R104 are 33 ohm series resistors in SWD. PC6 (pin 122) receives FPGA_DONE_MCU and PC7 (pin 123) receives FPGA_INIT_N_MCU through the SYS3V3-powered U901 return buffer.

MIRA		
Sheet: /MCU/100_MCU_POWER_CLOCKS_AND_DEBUG/		
File: 100_MCU_POWER_CLOCKS_AND_DEBUG.kicad_sch		
Title: STM32 supply, clocks, reset and debug		
Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 3/30

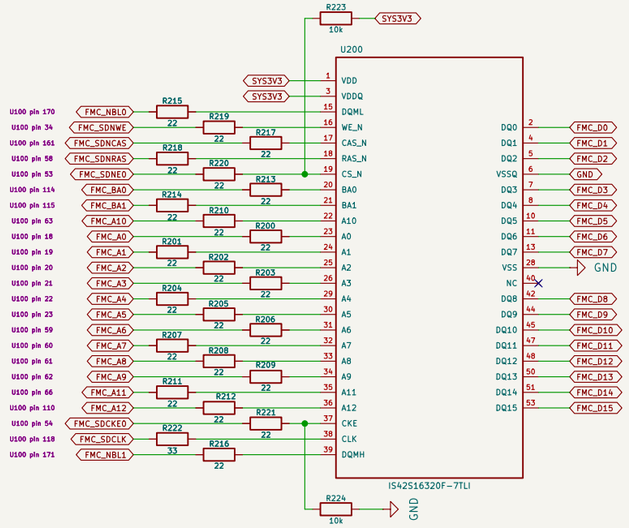
Main MCU FMC interface

Main MCU FMC interface | Rev.A



MAIN MCU SDRAM SOURCE DAMPING

Main MCU 64 MiB SDRAM | Rev.A



SDRAM supply bypass and reservoir



Memory path

The FMC unit of U100 drives U200, an IS42S16320F−7TLI SDRAM. The device provides 32M x 16 bits, or 64 MiB, of volatile storage. It belongs to the MCU address space and is separate from the FPGA DDR3L memory.

Bus organisation

FMC_D0 through FMC_D15 form the bidirectional data bus. FMC_A0 through FMC_A12 and FMC_BA0/FMC_BA1 select the row, column and internal bank according to the SDRAM command. FMC_NBL0 and FMC_NBL1 connect the two byte masks, allowing access to either half of the 16−bit word.

Commands and clock

FMC_SDNRAS, FMC_SDNCAS and FMC_SDNWE encode SDRAM commands. FMC_SDNE0 selects the device and FMC_SDCKE0 controls clock enable. FMC_SDCLK is routed through the separate 33 ohm R222. The other shown source−damping resistors are 22 ohms.

Startup bias and power

R223 pulls chip select high and R224 pulls clock enable low while the MCU pins are not actively driving them. U200 uses SYS3V3. Seven 100 nF capacitors and the 4.7 uF C207 provide local bypass and a supply reservoir.

Firmware and placement

Configure the FMC geometry, command sequence, timing parameters and refresh interval for this exact memory before application use. The schematic does not define those software values. Keep the clock, command and data routes controlled as a synchronous bus; retain the resistor−to−signal association shown here.

MIRA

Sheet: /MCU/200_MCU_FMC_AND_SDRAM/
File: 200_MCU_FMC_AND_SDRAM.kicad_sch

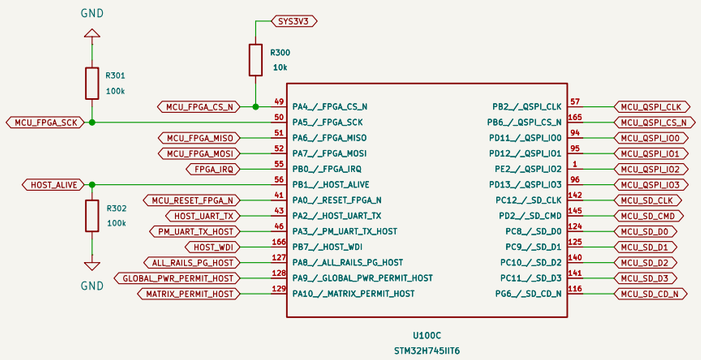
Title: STM32 FMC and 64 MiB SDRAM

Size: A3	Date: 2026−10−18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 4/30

MCU nonvolatile storage and board control

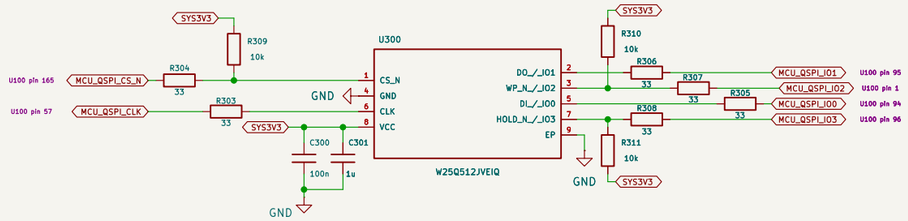
Main MCU serial memory and control

Main MCU serial memory and control | Rev.A



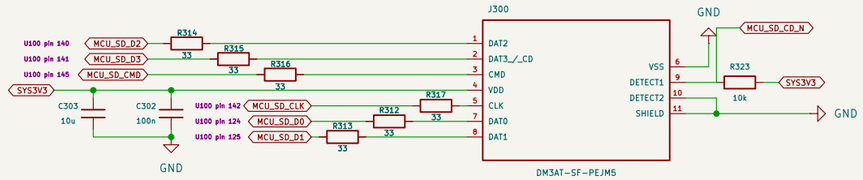
Main MCU 64 MiB QSPI NOR flash

Main MCU 64 MiB QSPI NOR flash | Rev.A

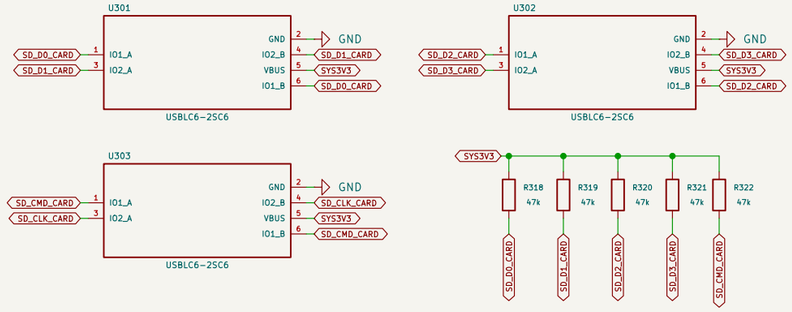


Main MCU microSD connector

Main MCU microSD connector | Rev.A



Main MCU microSD bias and ESD | Rev.A



QSPI program and data storage

U300 is a W25Q512JVEIQ NOR flash with 512 Mbit capacity, equivalent to 64 MiB. MCU_QSPI_CLK, MCU_QSPI_CS_N and four data signals connect it to U100. R303 through R308 are 33 ohm series resistors. The 10 kohm pull-ups keep chip select and the IO2/IO3 multifunction pins defined during startup.

Removable microSD storage

J300 connects a four-bit SD interface: clock, command and MCU_SD_D0 through MCU_SD_D3. The 33 ohm resistors separate MCU nets from SD_*_CARD nets. The 47 kohm resistors bias command and data. R323 pulls MCU_SD_CD_N high; the card-detect switch returns this net to ground. It reaches U100 PG6 (pin 116). C302 and C303 provide 100 nF and 10 uF local supply capacitance.

Connector protection

U301, U302 and U303 provide ESD paths for the six SD signal lines at the removable-card interface. The card shell and return contacts connect to the indicated ground network.

FPGA command interface

MCU_FPGA_SCK, MOSI, MISO and CS_N form the buffered command link, with FPGA_IRQ as an event return. MCU_RESET_FPGA_N reaches PROGRAM_B through the configuration-sheet open-drain buffer. FPGA_DONE_MCU and FPGA_INIT_N_MCU reach PC6/PC7 on the MCU supply sheet. Message formats and register maps must be implemented in firmware and FPGA logic.

Power-manager handshake

HOST_ALIVE, HOST_WDI and HOST_UART_TX communicate host activity and commands. ALL_RAILS_PG_HOST, GLOBAL_PWR_PERMIT_HOST, MATRIX_PERMIT_HOST and PM_UART_TX_HOST return status and permissions. These nets report the independent power manager state; they do not replace the hardware protection circuitry.

MIRA

Sheet: /MCU/300_MCU_QSPI_MICROSD_AND_CONTROL/
File: 300_MCU_QSPI_MICROSD_AND_CONTROL.kicad_sch

Title: MCU nonvolatile storage and board control

Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 5/30

INTERCONNECT — Power inputs and supervisory boundary

Consecutive circuit order; select a child sheet to inspect its circuit and sheet-specific explanation.

400_SYSTEM_POWER_AND_SUPERVISION_INTERFACE

Physical external rails, 3.3 V AON interface, Ioff status receivers and inhibit return.

Group scope

Explicit Control-side power and permission connector. Separate SYS3V3 and IO3V3 receivers prevent joining disabled receiver domains.

Reading guide

Each numeric child contains its own circuit explanation. Identical global labels connect between leaves. A shared U reference denotes one physical multi-unit device.

Physical references follow the first circuit leaf containing the component; multi-unit devices are not duplicated.

MIRA

Sheet: /INTERCONNECT/
File: INTERCONNECT.kicad_sch

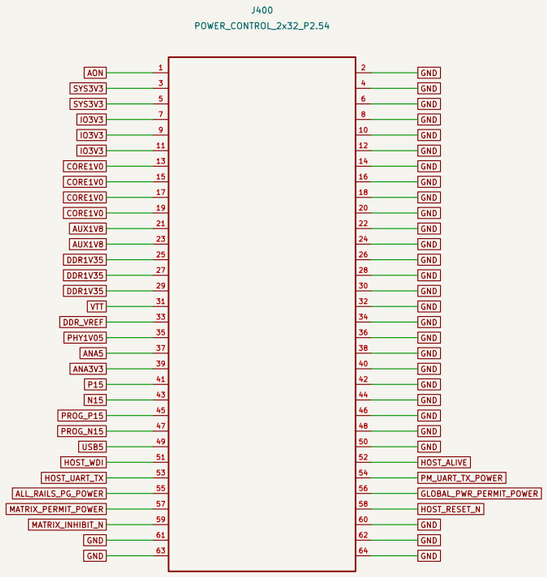
Title: Power inputs and supervisory boundary

Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 6/30

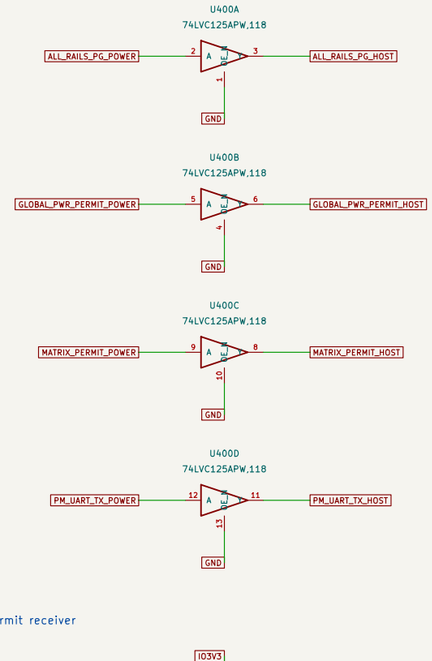
Power and supervision interface

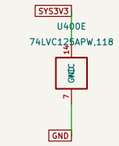
Control-side pin assignment; AON = 3.3 V. Match the mating Power board before PCB release.

J400 – rails and supervisor signals



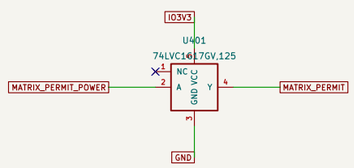
SYS3V3 receivers: powered-off isolation

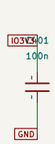






IO3V3 permit receiver





Shared Power permit is received independently in SYS3V3 and IO3V3; these output nets are not shorted.

Default states and open-drain supervision



Inputs from Power are 3.3 V logic. PG/permit default low; UART defaults high. HOST_RESET_N is driven only by an open-drain reset source. MATRIX_INHIBIT_N is an open-drain return to Power with an AON pull-up here. Watchdog/alive default low.

Power rails are physical inputs

J400 connects the external supplies to Control. Parallel CORE1V0, IO3V3, DDR1V35 and GND contacts provide an explicit route; their final current rating, mating height and return geometry must be checked against the Power board. A generic 2.54 mm 2x32 footprint defines this Control-side prototype interface.

Separate receiving domains

U400 uses SYS3V3 for the MCU-side power-good, permit and UART signals. U401 uses IO3V3 for the matrix logic. Both Nexperia devices specify Ioff, so an AON-powered supervisor does not drive a disabled receiver domain through its outputs.

Permission and hardware inhibit

MATRIX_PERMIT_POWER is the common supervisor permission. Its two buffered outputs remain separate nets. The matrix receiver is pulled low on sheet 1300. MAT_FAULT_Q produces MATRIX_INHIBIT_N through Q2203; the new connector now exposes that output to the Power supervisor.

Required supply contract

AON is 3.3 V and starts before SYS3V3/IO3V3. VTT must source and sink at DDR1V35/2; DDR_VREF is a quiet reference at the same nominal level. P15/N15 and PROG_P15/PROG_N15 are separate analog domains. Signal mapping is defined here, not inferred from a missing mating board.

MIRA		
Sheet: /INTERCONNECT/400_SYSTEM_POWER_AND_SUPERVISION_INTERFACE/		
File: 400_SYSTEM_POWER_AND_SUPERVISION_INTERFACE.kicad_sch		
Title: Power and supervision interface		
Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 7/30

FPGA — Programmable logic, memory and external links

Consecutive circuit order; select a child sheet to inspect its circuit and sheet-specific explanation.

500_FPGA_POWER_AND_DECOUPLING

One Artix-7 package: core, auxiliary and selected I/O supply decoupling.

600_FPGA_CONFIGURATION_AND_XADC

Master SPI flash, JTAG, PROGRAM_B recovery and XADC supply filter.

700_FPGA_USB3_FIFO_INTERFACE

USB Type-C orientation, ESD, FT601 FIFO and isolated VBUS detection.

800_FPGA_DDR3L_AND_REFERENCE_CLOCK

FPGA DDR3L data groups, terminations and AC-coupled 200 MHz reference.

900_FPGA_ETHERNET_AND_MCU_INTERFACE

Ethernet PHY/magnetics and domain-powered MCU/FPGA control buffers.

Group scope

Artix-7 supplies, configuration, USB FIFO, DDR3L, Ethernet PHY and buffered MCU link. All FPGA sections belong to one physical package.

Reading guide

Each numeric child contains its own circuit explanation. Identical global labels connect between leaves. A shared U reference denotes one physical multi-unit device.

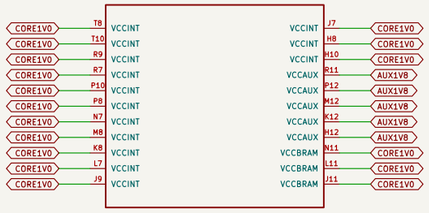
Physical references follow the first circuit leaf containing the component; multi-unit devices are not duplicated.

MIRA	
Sheet: /FPGA/	
File: FPGA.kicad_sch	
Title: Programmable logic, memory and external links	
Size: A3	Date:
KiCad E.D.A. 10.0.5	**Rev: B**
	Id: 8/30

FPGA supply domains and local bypass

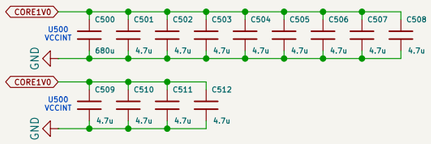
U500A

FPGA core supply

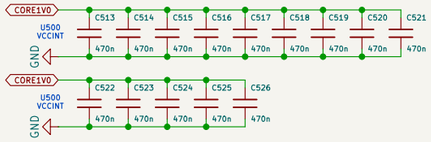


XC7A200T-2FBG484I

VCCINT local capacitors; C500 bulk positive to CORE1V0.



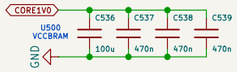
VCCINT 470nF bank; place near the FPGA supply.



Local VCCAUX capacitors

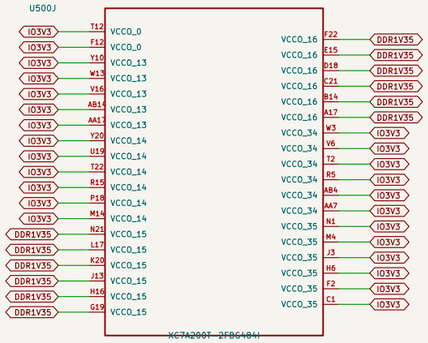


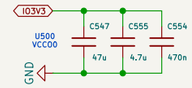
Local VCCBRAM C536..C539 capacitors

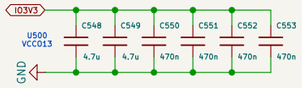


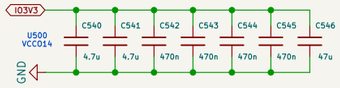
FPGA bank supply

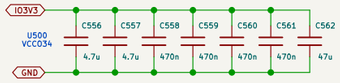
U500J

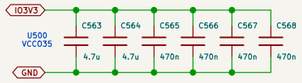


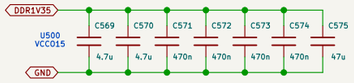


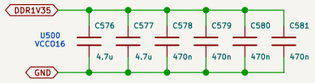








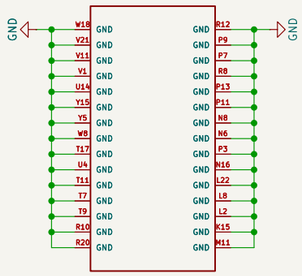




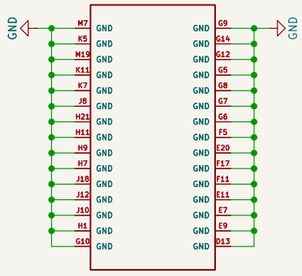
FPGA ground

U500K

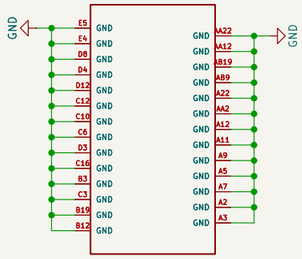
XC7A200T package ground balls are shown explicitly.



U500L

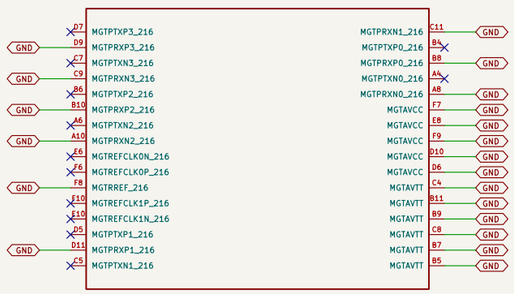


U500M



Unused FPGA GTP transceiver quad

U500I



Unused-quad connections follow AMD UG482 table 5-4.

One FPGA, several supply groups

All U500 units shown here belong to the XC7A200T-2FBG484I. CORE1V0 feeds VCCINT and VCCBRAM; AUX1V8 feeds VCCAUX. These rails serve different internal circuits and remain separate even though their return is the common GND network.

Core and block-RAM reservoirs

C500 is the 680 uF bulk capacitor on CORE1V0. The surrounding 4.7 uF and 470 nF banks provide distributed bypass for the VCCINT supply balls. The VCCBRAM group includes the 100 uF C536 and its local 470 nF capacitors. The drawing groups these parts electrically; their final positions must follow the package supply balls.

I/O bank voltages

VCCO0, VCCO13, VCCO14, VCCO34 and VCCO35 connect to IO3V3. VCCO15 and VCCO16 connect to DDR1V35 for the DDR3L interface. Each bank has its own nearby capacitor group, although several banks share the same upstream rail.

Bank assignment matters

The bank voltage determines which I/O standards can be used on its signals. Configuration, Ethernet, USB FIFO and matrix-control signals must follow their actual bank assignments. The DDR banks must retain the DDR1V35 supply and the reference connections shown on the DDR3L sheet.

Ground and unused transceivers

The package ground unit preserves all physical ground balls. The unused GTP quad is shown separately with its existing grounded and open pins; it is not the USB3 data path. USB communication uses an external FT601 and a parallel FPGA FIFO interface.

MIRA

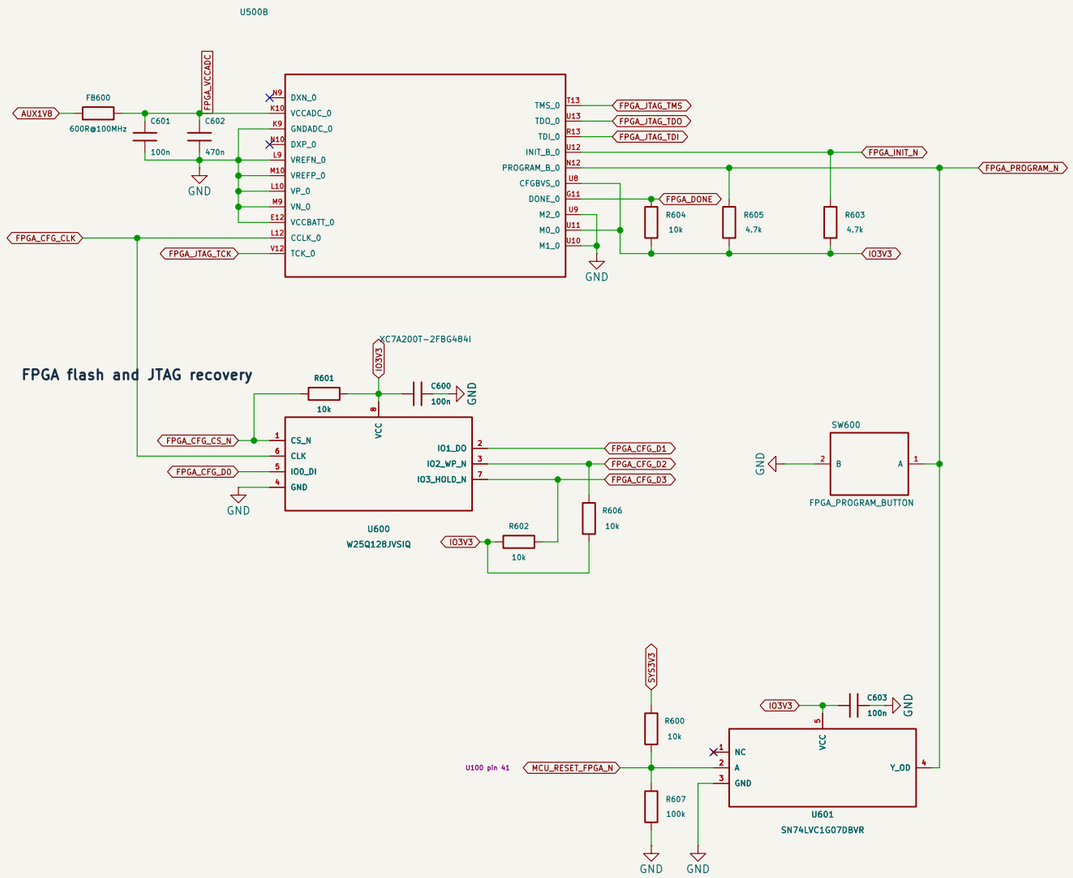
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File: 500_FPGA_POWER_AND_DECOUPLING.kicad_sch

Title: FPGA supply domains and local bypass

Size: A3	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.5		Id: 9/30

FPGA configuration, recovery and XADC supply

FPGA configuration and XADC



Configuration storage

U600 is the W25Q128JVSIQ configuration flash. FPGA_CFG_CLK, FPGA_CFG_CS_N and FPGA_CFG_D0 through FPGA_CFG_D3 connect it to U500. The flash and configuration bank use IO3V3. R601, R602 and R606 provide 10 kohm pull-ups on select and the flash multifunction data pins.

Mode and status pins

M[2:0] = 001 selects Master SPI; CFGBVS connects to IO3V3. R603 and R605 are 4.7 kohm pull-ups on INIT_N and PROGRAM_N; R604 is 10 kohm on DONE. FPGA_DONE and FPGA_INIT_N pass through U901 to the MCU PC6/PC7 inputs. The status signals indicate configuration state; they do not prove application logic, DDR calibration or communication readiness.

Manual and MCU recovery

SW600 pulls FPGA_PROGRAM_N low. U601 is an SN74LVC1G07 open-drain buffer driven by MCU_RESET_FPGA_N: low asserts PROGRAM_B; high releases it to R605. R600/R607 bias the input and C603 bypasses the buffer. A low pulse followed by release restarts configuration after power-up. Holding PROGRAM_B low from initial power-on does not delay first configuration.

JTAG service access

J600 exposes FPGA_JTAG_TCK, TMS, TDI and TDO, together with the indicated supply reference and return. This path provides direct programming and recovery access independently of the normal flash-loading path.

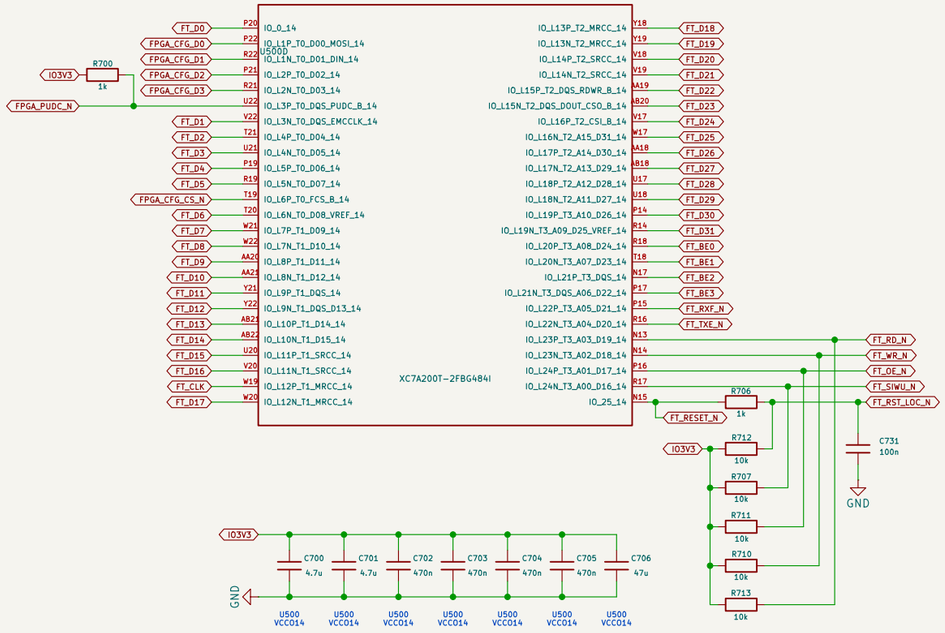
XADC analog supply

FB600 filters AUX1V8 into the FPGA VCCADC pin. C601 and C602 provide 100 nF and 470 nF local bypass. The shown external analog input and reference pins are grounded. XADC internal monitoring, when used, is selected and read by FPGA logic or firmware; this sheet contains its supply connections.

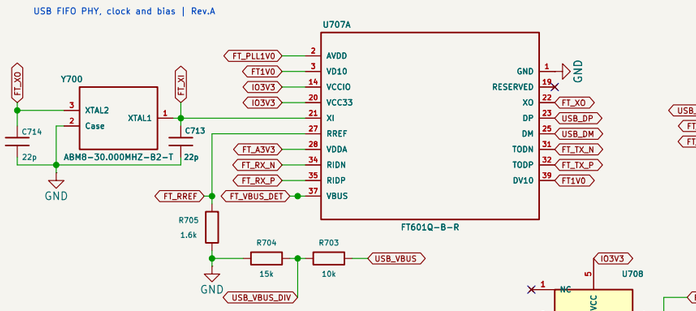
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Sheet: /FPGA/600_FPGA_CONFIGURATION_AND_XADC/	
File: 600_FPGA_CONFIGURATION_AND_XADC.kicad_sch	
Title: FPGA configuration, recovery and XADC supply	
Size: A3 / Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5	Id: 10/30

USB Type-C and 32-bit FPGA FIFO

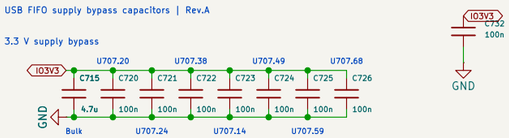
FPGA USB FIFO and configuration I/O



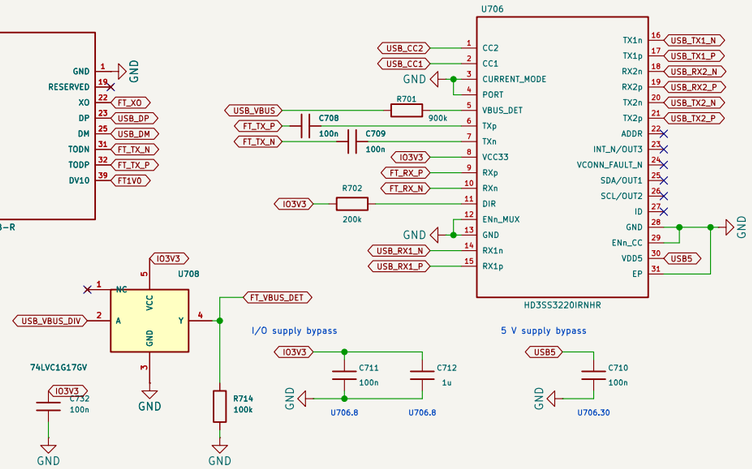
USB FIFO PHY, clock and bias



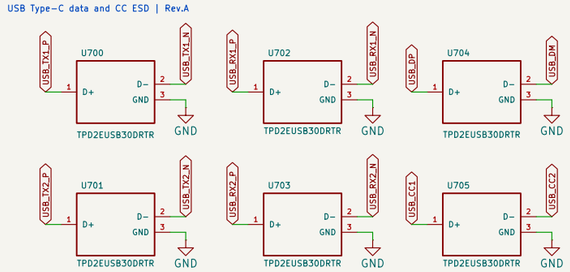
USB FIFO supply bypass capacitors



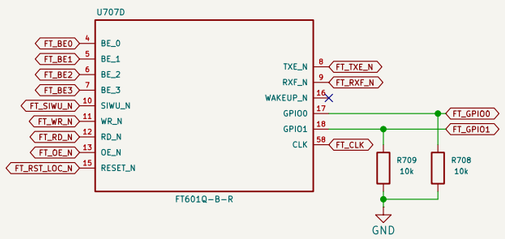
USB CC and SuperSpeed orientation



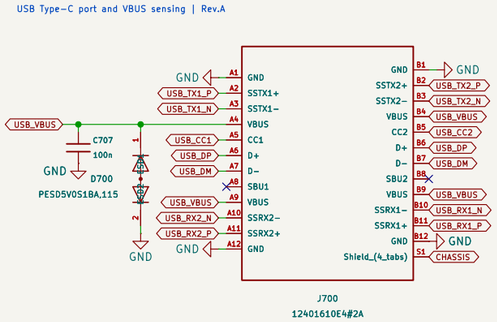
USB Type-C data and CC ESD



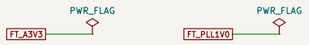
USB Type-C port and VBUS sensing



USB Type-C port and VBUS sensing



Filtered supply: source declaration for ERC



Connector-to-FIFO path

J700 is the USB Type-C port. U706, the HD3SS3220, handles CC detection and SuperSpeed lane orientation. U707, the FT601Q-B-R, terminates the USB link and presents a parallel FIFO to FPGA bank 14. USB_DP and USB_DM connect the USB2 pair directly to the FT601 interface.

FPGA data and handshake

FT_D0 through FT_D31 form the 32-bit bidirectional data bus; FT_BE0 through FT_BE3 identify valid bytes. FT_CLK provides the interface clock. RXF_N and TXE_N report FIFO availability, while RD_N, WR_N and OE_N control transfers. SIWU_N and the GPIO signals are also brought into the control network.

Defined startup conditions

The 10 kohm pull resistors hold read, write and output-enable controls inactive. FT_RST_LOC_N has a pull-up and receives FT_RESET_N through 1 kohm R706. FT_GPIO0/1 have pull-downs. R700 is a 1 kohm pull-up on FPGA_PUDC_N, meeting the dedicated pin tie requirement. Configuration data pins sharing this bank retain their named flash connections.

Clock, sensing and protection

Y700 uses a 30 MHz crystal and two 22 pF capacitors; R705 is 1.6 kohm. R703 (10 kohm) and R704 (15 kohm) form a 3.0 V nominal VBUS divider. U708, a 74LVC1G17GV Schmitt buffer on IO3V3, drives FT_VBUS_DET and isolates the FT601 when IO3V3 is off. C732 bypasses U708; R714 holds its output low when unpowered. Data/CC protection remains at the port.

Separate power domains

IO3V3 supplies FIFO I/O and 3.3 V pins. FT1V0, FT_PLL1V0 and FT_A3V3 have separate local bypass, with ferrite filtering for the analog groups. USB5 supplies the Type-C controller. USB_VBUS and USB5 remain distinct named nets in the schematic.

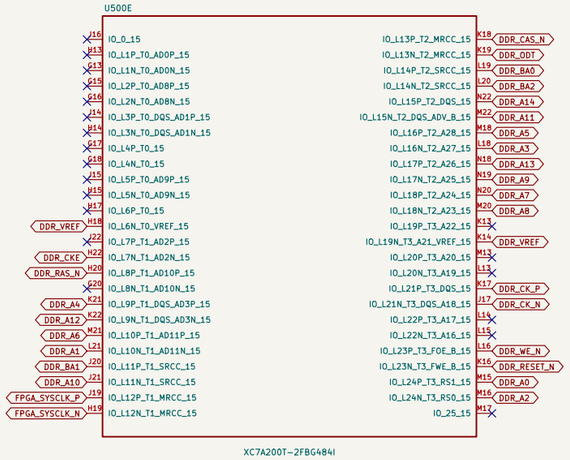
MIRA		
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File: 700_FPGA_USB3_FIFO_INTERFACE.kicad_sch		
Title: USB Type-C and 32-bit FPGA FIFO		
Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 11/30

FPGA DDR3L memory and 200 MHz reference

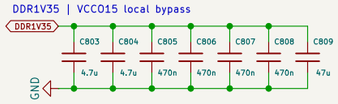
FPGA DDR3L address, command and clock

FPGA DDR3L address and command

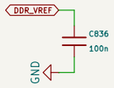
FPGA bank15 | command / clocks (E)



DDR1V35 | VCCO15 local bypass

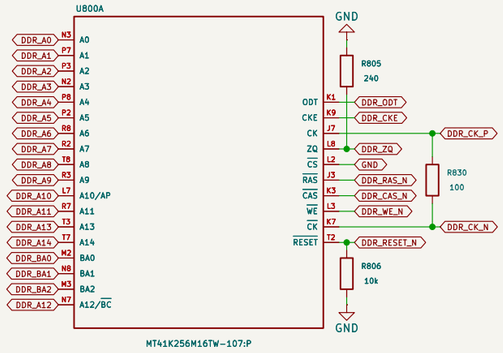


DDR_VREF | bank15 H18 / K14



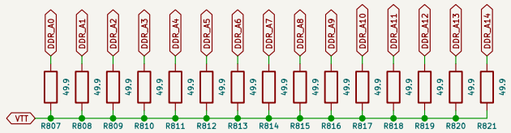
DDR3L address and command pins

DDR3L address / command (A)

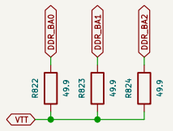


DDR3L signal termination

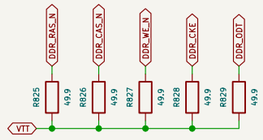
VTT | A0..A5



VTT | A12..A14 and BA0..BA2

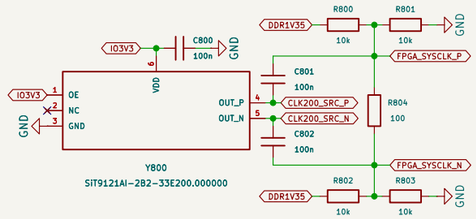


VTT | RAS / CAS / WE / CKE / ODT



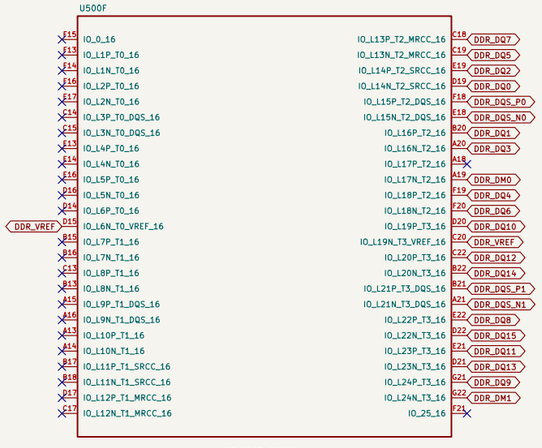
FPGA 200MHz system clock

FPGA 200MHz reference | SYSCLK J19 / H19

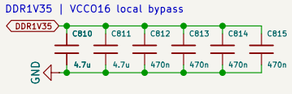


DDR3L address and command pins

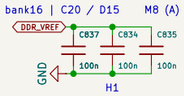
FPGA bank16 | data byte lanes (F)



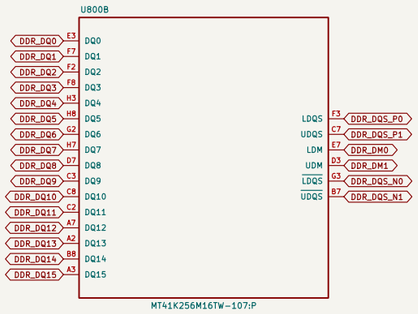
DDR1V35 | VCCO16 local bypass



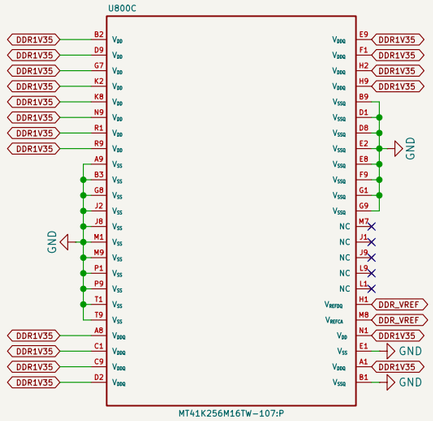
VREF bank16 | C20 / D15 M8 (A)



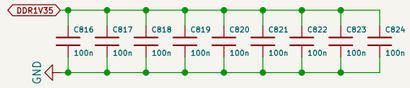
DDR3L DQ / DQS / DM (B)



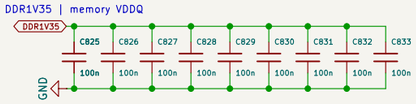
DDR3L supply / ground balls (C)



DDR1V35 | memory VDD



DDR1V35 | memory VDDQ



Dedicated FPGA memory

U800 is an MT41K256M16TW-107:P DDR3L device, organised as 256M x 16 bits, or 512 MiB. FPGA bank 15 carries address, command and memory clock signals; bank 16 carries the two data byte lanes. This memory is independent of the MCU FMC SDRAM.

Address and data lanes

DDR_A0 through DDR_A14 and DDR_BA0 through DDR_BA2 address the memory. RAS_N, CAS_N, WE_N, CKE, ODT and RESET_N provide command and state control. DDR_DQ0 through DDR_DQ15 form the data bus. Each eight-bit lane has a differential DQS strobe and its own DM mask.

Termination and calibration

The 49.9 ohm resistor groups terminate the drawn address and control signals to VTT. R830 provides 100 ohms across DDR_CK_P/N. R805 is the 240 ohm ZQ calibration resistor to ground; R806 pulls DDR_RESET_N low. The memory chip-select pin is grounded in the shown single-device arrangement.

Memory power and references

DDR1V35 supplies memory VDD/VDDQ and FPGA VCCO15/VCCO16. DDR_VREF reaches both memory reference pins and the indicated bank reference balls. The dedicated 100 nF, 470 nF, 4.7 uF and 47 uF capacitor groups support these domains locally.

Reference clock and implementation

Y800 provides a 200 MHz differential reference. C801 and C802 AC-couple its outputs; four 10 kohm resistors bias the FPGA inputs from DDR1V35, and R804 terminates the pair with 100 ohms. FPGA memory-controller configuration and PCB routing must match the actual pin map, memory device and clock.

MIRA

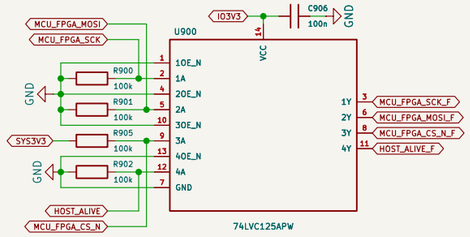
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Title: FPGA DDR3L memory and 200 MHz reference

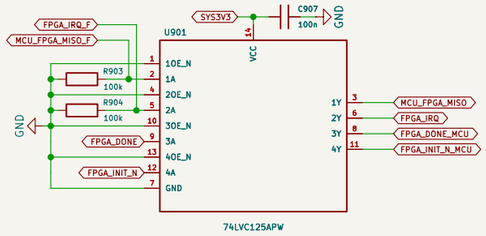
Size: A3	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.5		Id: 12/30

Ethernet physical interface and MCU buffers

MCU to FPGA domain buffers

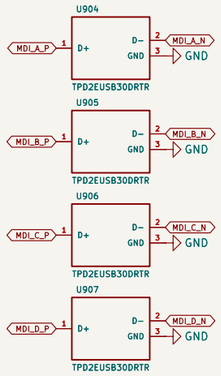


FPGA to MCU domain buffers



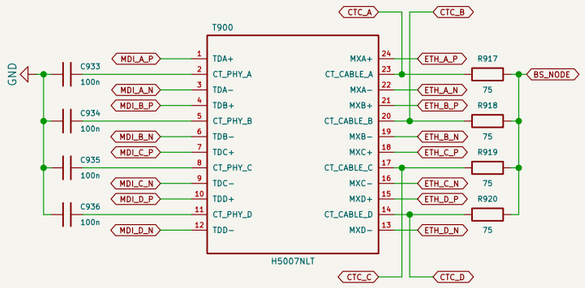
Ethernet RJ45, ESD and chassis

Ethernet RJ45, ESD and chassis | Rev.A

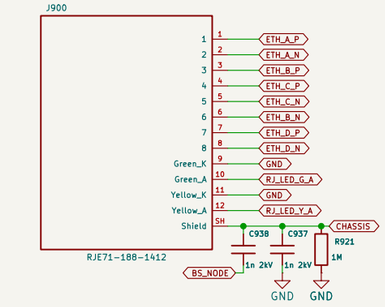


Ethernet magnetics and center taps

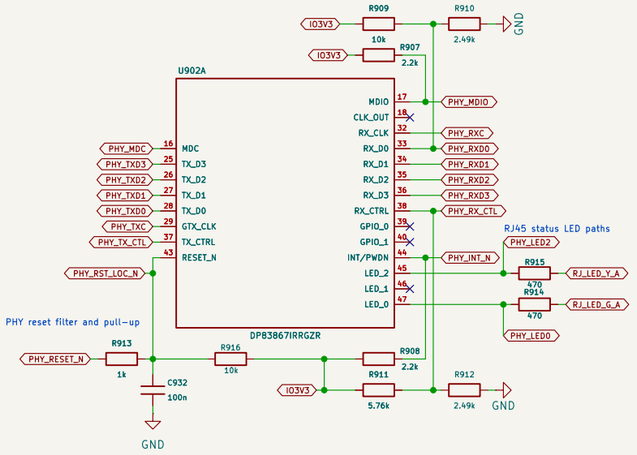
Ethernet magnetics and center taps | Rev.A



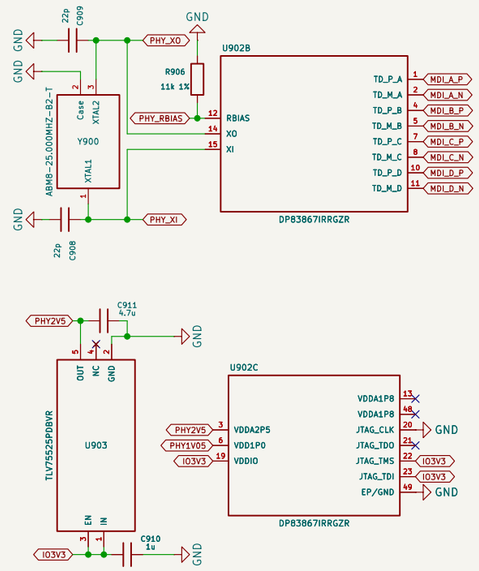
Ethernet RJ45, ESD and chassis



Ethernet RGMII and boot straps

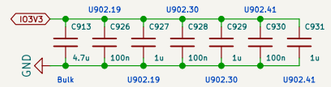


Ethernet PHY supply and clock

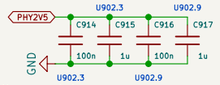


Ethernet PHY supply bypass capacitors | Rev.A

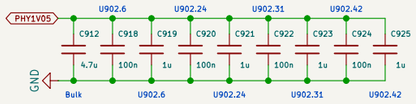
3.3 V I/O bypass



2.5 V analog supply bypass



1.05 V core supply bypass



Ethernet signal path

FPGA bank 13 connects by RGMII to U902, the DP83867IRRGZR PHY. Four MDI differential pairs pass through T900 magnetics to J900 RJ45. MDC/MDIO configure and read the PHY; PHY_INT_N reports events. Ethernet framing and network protocols are implemented by the FPGA design and software.

PHY clock, reset and straps

Y900 is a 25 MHz crystal with 22 pF capacitors. R906 is the 11 kohm bias resistor. Resistor pairs on PHY_RXD0 and PHY_RX_CTL establish startup strap levels. PHY_RESET_N reaches the local reset node through 1 kohm R913; R916 and C932 provide pull-up and filtering.

PHY supply domains

IO3V3 powers the I/O pins. U903 generates PHY2V5 from IO3V3 for the analog supply. PHY1V05 is a separate incoming core rail. The grouped 100 nF, 1 uF and 4.7 uF capacitors bypass the corresponding physical supply pins.

Cable interface and chassis

U904 through U907 protect the four Ethernet pairs. Four 75 ohm resistors join cable-side centre taps to BS_NODE, which couples to CHASSIS through C938. R921 and C937 connect chassis to circuit ground. R914 and R915 limit the RJ45 status LED currents.

Buffered MCU link

U900 forwards SCK, MOSI, CS_N and HOST_ALIVE, powered by IO3V3. U901 returns MISO, FPGA_IRQ, FPGA_DONE and FPGA_INIT_N, powered by SYS3V3 with C907. Both are Nexperia 74LVC125APW,118 with specified ioff. U901 3Y/4Y drive MCU PC6/PC7 through FPGA_DONE_MCU/FPGA_INIT_N_MCU. Output enables remain grounded; 100 kohm resistors define startup levels.

MIRA
Sheet: /FPGA/900_FPGA_ETHERNET_AND_MCU_INTERFACE/
File: 900_FPGA_ETHERNET_AND_MCU_INTERFACE.kicad_sch
Title: Ethernet physical interface and MCU buffers
Size: A3 | Date: 2026-10-18 | Rev: B
KiCad E.D.A. 10.0.5 | Id: 13/30

DAC — Analog commands and source buffers

Consecutive circuit order; select a child sheet to inspect its circuit and sheet-specific explanation.

1000_DAC_BL0_31_AND_BUFFERS

Thirty-two DAC setpoints and bipolar analog output buffers for BL sources.

1100_DAC_SERVICE_AND_FORMING

Service voltage and current command DACs; READ shares the programmable ILIM_SET target. Commands no longer power DUT directly.

1200_DAC_FPGA_CONTROL_INTERFACE

FPGA serial controls for the DAC banks and defined inactive states.

Group scope

Thirty-two BL voltages plus service voltage and current commands. The shared source-side CV/CC regulator is on sheet 2100; polarity is selected by matrix routing.

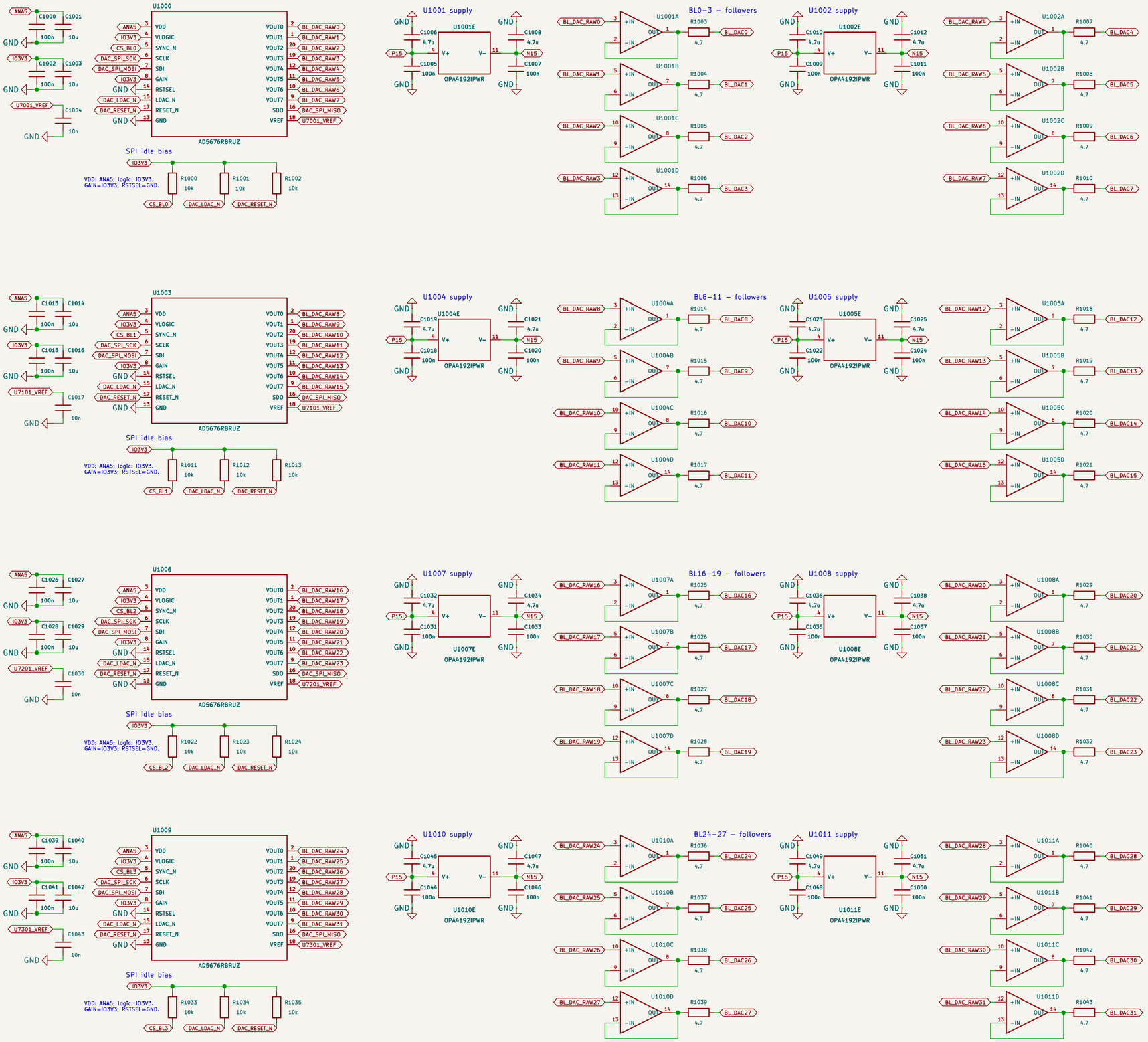
Reading guide

Each numeric child contains its own circuit explanation. Identical global labels connect between leaves. A shared U reference denotes one physical multi-unit device.

Physical references follow the first circuit leaf containing the component; multi-unit devices are not duplicated.

MIRA	
Sheet: /DAC/	
File: DAC.kicad_sch	
Title: Analog commands and source buffers	
Size: A3	Date:
KiCad E.D.A. 10.0.5	**Rev: B**
	Id: 14/30

Thirty-two programmable bit-line voltages



Channel allocation

Four AD5676R DACs provide all 32 bit-line setpoints. U1000 serves channels 0–7, U1003 serves 8–15, U1006 serves 16–23 and U1009 serves 24–31. Each device has its own CS_BL0–3 selection and shares the serial bus, LDAC and RESET controls.

Analog and logic supplies

The DAC analog supply is ANA5, while its logic interface uses IO3V3. GAIN is tied to IO3V3 and RSTSEL to GND on each DAC. The reference pins have local bypass capacitors. Firmware must use these fixed hardware settings when converting a requested voltage into a DAC code.

Buffered output path

BL_DAC_RAW0–31 feed the noninverting inputs of the OPA4192 followers. Two quad amplifiers serve each eight-channel group. Their feedback is local, and each output reaches its BL_DAC net through a 4.7 ohm series resistor. These resistors separate the amplifier output from the downstream switch network.

Connection to the crossbar

BL_DAC0–31 are voltage sources for the bit-line routing circuit, rather than permanent connections to the socket. The corresponding MAT_BL_VMM mask determines which buffered source is connected. A disabled switch allows the same bit line to be grounded or used by the service route.

Local decoupling

The amplifier supply units use P15 and N15 with local 100n and 4.7u capacitors. DAC supply bypassing is grouped beside each converter. Keep these associations when placing the PCB so that every package has a short supply return.

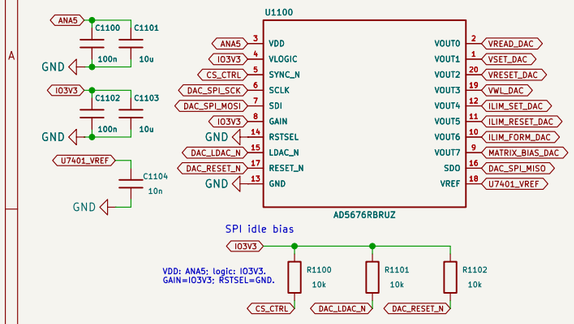
MIRA

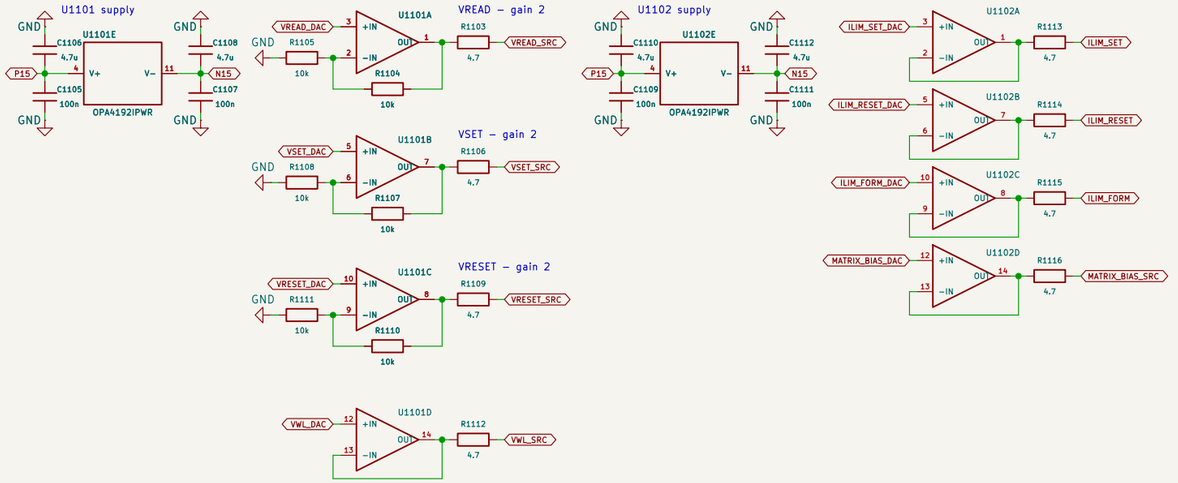
Sheet: /DAC/1000_DAC_BL0_31_AND_BUFFERS/
File: 1000_DAC_BL0_31_AND_BUFFERS.kicad_sch

Title: Thirty-two programmable bit-line voltages

Size: A3	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.5		Id: 15/30

Service voltages, current limits and forming source





Eight service setpoints

U1100, an AD5676R selected by CS_CTRL, generates VREAD_DAC, VSET_DAC, VRESET_DAC and VWL_DAC on outputs 0–3. Outputs 4–6 generate ILIM_SET_DAC, ILIM_RESET_DAC and ILIM_FORM_DAC. Output 7 generates MATRIX_BIAS_DAC. All eight settings therefore belong to one converter but serve different downstream functions.

Voltage gain and distribution

U1101 channels A–C amplify VREAD_DAC, VSET_DAC and VRESET_DAC with equal 10k feedback and ground resistors: the nominal noninverting gain is 2. Channel D buffers VWL_DAC at unity gain. U1102 buffers the three current-limit commands and MATRIX_BIAS_DAC at unity gain. All nine source interfaces include 4.7 ohm series output resistors.

Separate forming path

U1103 generates VFORM_DAC_RAW; U1104A buffers it to VFORM_SRC.
U1104B powers the same OPA192 from PROG_P15/PROG_N15 (pins 5/2).
Configure FORMING to unipolar 0...+10 V for the shared CV/CC source.
Select the DUT voltage sign with MAT_POLARITY, not a bipolar DAC range.

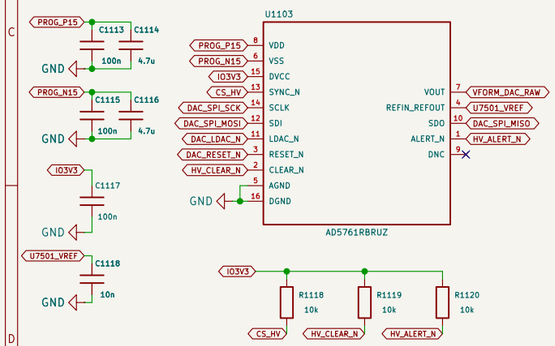
Digital controls

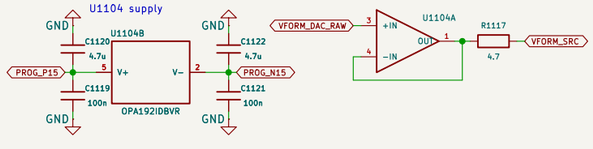
Both DACs share DAC_SPI_SCK, DAC_SPI_MOSI, DAC_SPI_MISO, DAC_LDAC_N and DAC_RESET_N. HV_CLEAR_N and HV_ALERT_N belong specifically to the forming DAC. U1100 has GAIN tied to IO3V3 and RSTSEL tied to GND.

Where the settings are used

U1601 selects the high-impedance voltage target MAT_VTARGET. The CV/CC source on sheet 2100 supplies DUT power. SRC also selects matching ILIM: READ and SET share ILIM_SET; RESET and FORM use their own current DAC outputs. Changing a service DAC during a WRITE pulse disables RUN and records a fault.

U1103 – forming DAC





WRITE sources: 0...+10 V only.
Use MAT_POLARITY for the voltage sign across the DUT.
Voltage targets feed the shared CV/CC source on sheet 2100.
READ shares ILIM_SET; reload current targets with outputs disabled.

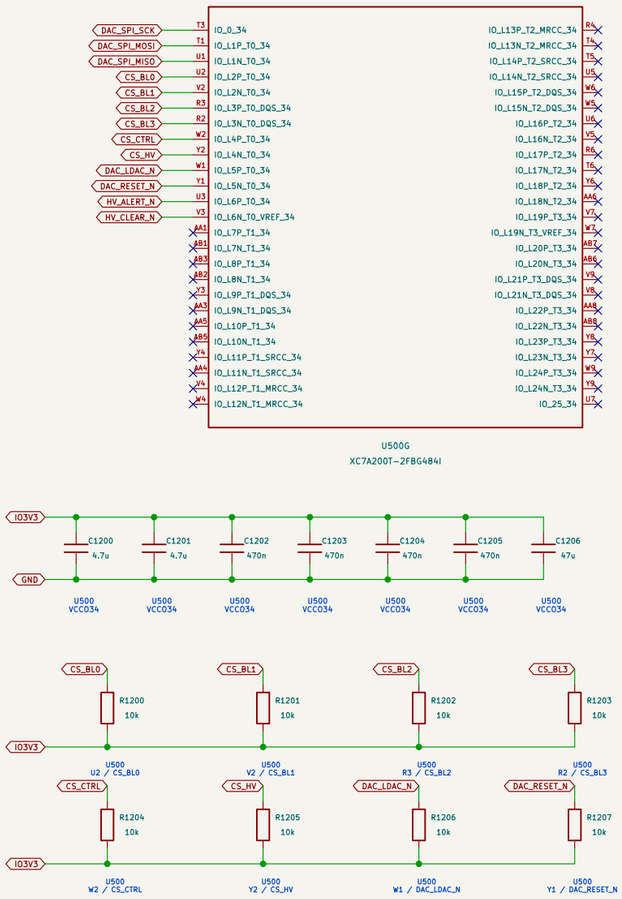
MIRA

Sheet: /DAC/1100_DAC_SERVICE_AND_FORMING/
File: 1100_DAC_SERVICE_AND_FORMING.kicad_sch

Title: Service voltages, current limits and forming source

Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 16/30

FPGA DAC timing interface



Shared serial bus

This section of U500 drives DAC_SPI_SCK and DAC_SPI_MOSI and receives DAC_SPI_MISO. The bus reaches four bit-line DACs, the service DAC and the forming DAC. Select only the intended device during each transfer; its output data share the same return wire.

Individual device selection

CS_BL0, CS_BL1, CS_BL2 and CS_BL3 select the four groups of eight bit-line outputs. CS_CTRL selects the service voltages and current-limit references. CS_HV selects the separate AD5761R forming source. The 10k bias resistors keep the select lines inactive when the FPGA pins are not driving.

Output update and recovery

DAC_LDAC_N is distributed to the DACs for coordinated output updates. DAC_RESET_N provides a common hardware reset. Their exact effect on output codes also depends on the DAC configuration and reset selection; firmware must establish known setpoints before enabling the matrix routing.

Forming status and clear

HV_ALERT_N returns the forming DAC status to the FPGA. HV_CLEAR_N is a separate forming-DAC clear control. These signals serve the DAC interface and do not replace MATRIX_PERMIT or the matrix fault latch.

Bank supply

The illustrated FPGA bank uses IO3V3. Its local 47u, 4.7u and 470n capacitors support the bank supply; the analog DAC and amplifier supplies are shown on their own circuit sheets.

MIRA

Sheet: /DAC/1200_DAC_FPGA_CONTROL_INTERFACE/
File: 1200_DAC_FPGA_CONTROL_INTERFACE.kicad_sch

Title: FPGA control of the DAC bank

Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 17/30

MATRIX — Held routing, line drivers and crossbar socket

Consecutive circuit order; select a child sheet to inspect its circuit and sheet-specific explanation.

1300_MATRIX_FPGA_CONTROL_AND_INTERLOCKS

Capture mode and route, compare active requests, pair voltage/current selection and qualify RUN; DAC updates during WRITE trip.

1400_MATRIX_SERIAL_CONTROL_REGISTERS

80-bit serial image; commit only while outputs are disabled; captured address fields and frame-valid arm contract.

1500_MATRIX_BL0_31_SWITCHING

BL0-31 switch paths; BL6/7 source identities corrected; return bleeders DNP.

1600_MATRIX_SERVICE_SOURCE_BANK_AND_POLARITY_ROUTING

High-impedance voltage command, service/polarity/return routing, upstream source OFF-to-GND and complete PI reset.

1700_MATRIX_WL0_7_DRIVERS_AND_NL0_7_CLAMPS_SELECTOR

WL drivers and NL clamp/return selection; return bleeders DNP.

1800_MATRIX_CROSSBAR_SOCKET_AND_BIAS

64-contact crossbar socket, case/substrate adapter nets and DNP options.

Physical references follow the first circuit leaf containing the component; multi-unit devices are not duplicated.

Group scope

Captured mode and address fields, 80-bit image, service bank, polarity and READ/WRITE return paths. Any active request mismatch inhibits RUN and raises configuration trip.

Reading guide

Each numeric child contains its own circuit explanation. Identical global labels connect between leaves. A shared U reference denotes one physical multi-unit device.

MIRA

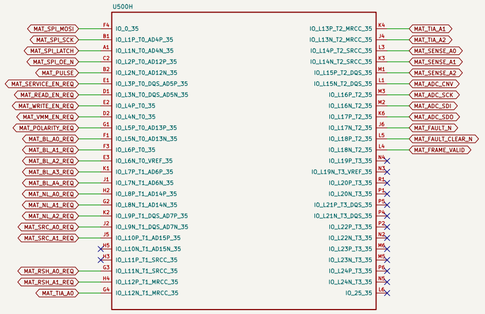
Sheet: /MATRIX/
File: MATRIX.kicad_sch

Title: Held routing, line drivers and crossbar socket

Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 18/30

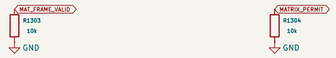
Matrix commands and hardware interlocks

FPGA MATRIX ENGINE / BANK 35

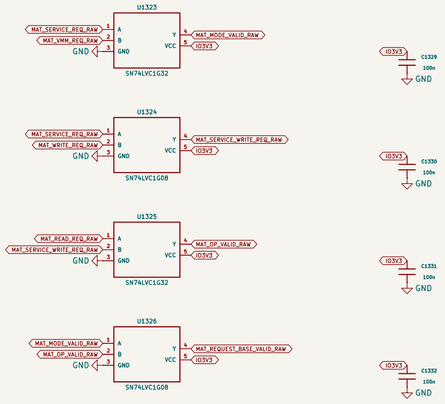




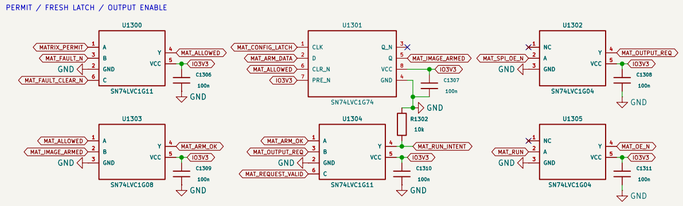
Fail-safe external permit / FPGA frame validator



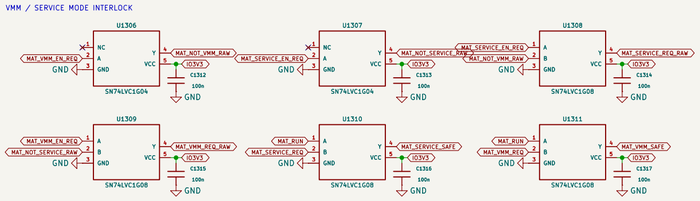
STRICT COMMAND VALIDATION



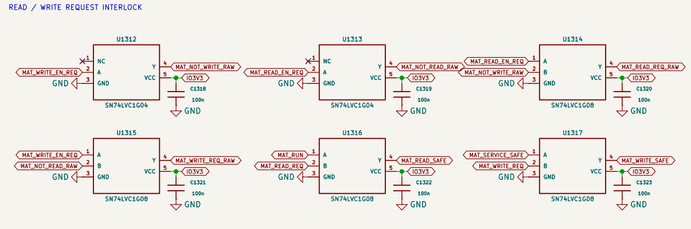
PERMIT / FRESH LATCH / OUTPUT ENABLE



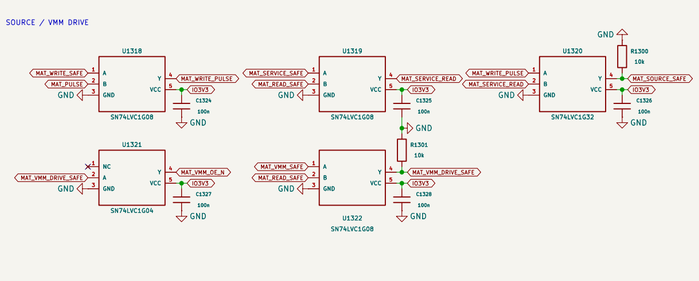
VMM / SERVICE MODE INTERLOCK



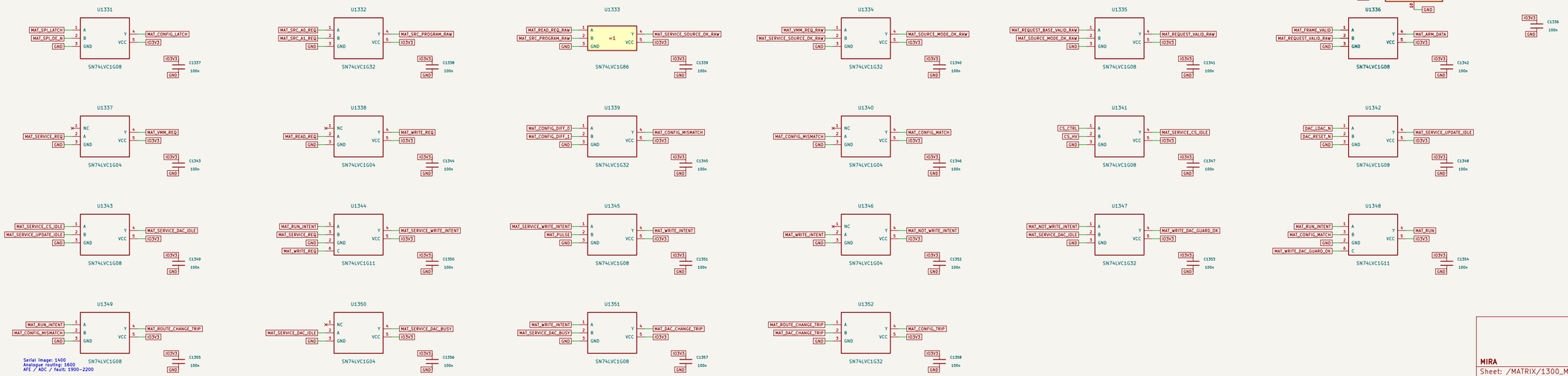
READ / WRITE REQUEST INTERLOCK



SOURCE / VMM DRIVE

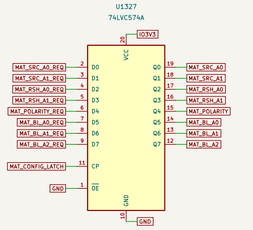


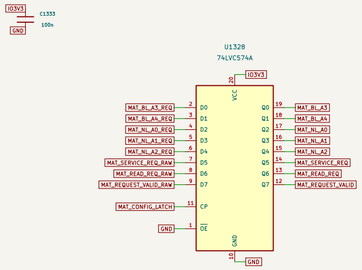
Source qualification / write-DAC guard / fault event

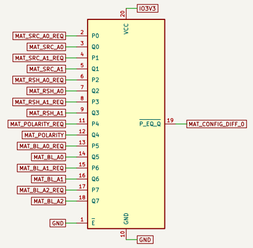


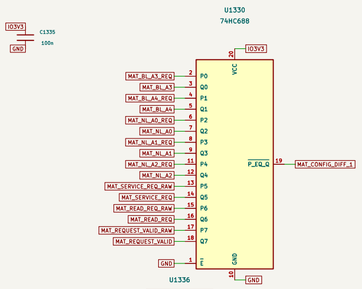
Serial Image: 1400
Analogue routing: 1600
AFE / ADC / fault: 1900–2200

Atomic route capture / live-request comparison









Power permission and safe defaults

MAT_ALLOWED requires external MATRIX_PERMIT, released fault clear and no latched fault. Power loss or clear resets frame arm. Capture registers have no defined startup state; their outputs cannot energize the matrix before a fresh qualified commit.

One atomic configuration commit

MAT_CONFIG_LATCH = MAT_SPI_LATCH & MAT_SPI_OE_N. Only a commit with outputs disabled clocks the 80-bit mask chain, both 8-bit configuration registers and frame arm. Keep LATCH low during OE changes. Hold all inputs stable around the qualified edge.

Routes held during operation

Two 74LVC574 registers capture SRC, RSH, POLARITY, BL and NL addresses plus exclusive service/read mode and request validity. Analog route controls use captured outputs. Two 74HC688 comparators detect request changes: mismatch blocks RUN and generates a latched fault request while output is intended.

Matched operation and source

SERVICE_READ requires source code 00. SERVICE_WRITE requires 01 SET, 10 RESET or 11 FORM. VMM_READ does not use the service-source code. The source and current-limit multiplexers share captured SRC bits. READ forces the analog programming range to its defined low-ohm setting.

Write command protection

CS_CTRL, CS_HV, DAC_LDAC_N and DAC_RESET_N must remain high throughout an intended WRITE pulse. A change suppresses the applied window and raises MAT_CONFIG_TRIP. VMM DAC chip selects are not included. READ may intentionally sweep its source DAC while the route remains fixed.

FPGA and serial-frame boundary

MAT_FRAME_VALID remains an HDL assertion: there is no external 80-clock counter, CRC or per-line mask checker. Validate exact frame length, no VMM/ground overlap, legal selected-line clamps and suitable WL selection. Address freeze does not validate the crossbar image.

Commit timing and re-arm

Use at least 1 us between the qualified LATCH edge and OE_N falling for the digital paths here; analog settling is additional and range dependent. A fault requires source zero, explicit clear release and a new full frame plus qualified commit. Never change OE while LATCH is high.

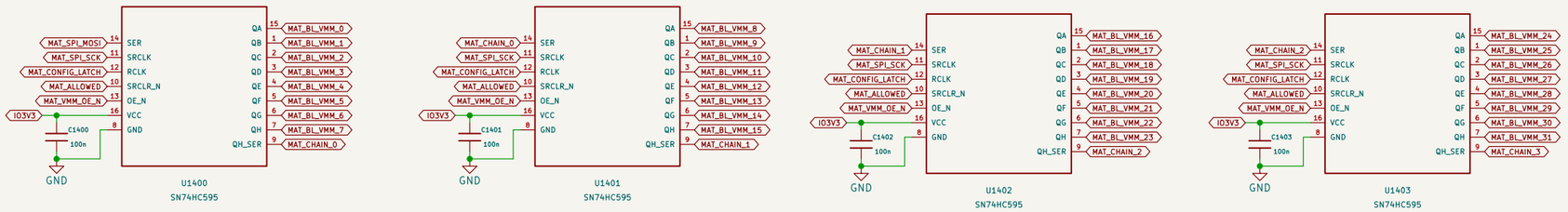
Stored energy and pulse timing

MAT_SOURCE_SAFE is SERVICE_READ or qualified WRITE_PULSE. It now controls the source actuator, not a charged pulse switch after current sense. Matrix capacitance and switch injection remain in the energy budget. FPGA bounds pulse duration and enforces zero-source discharge before rerouting.

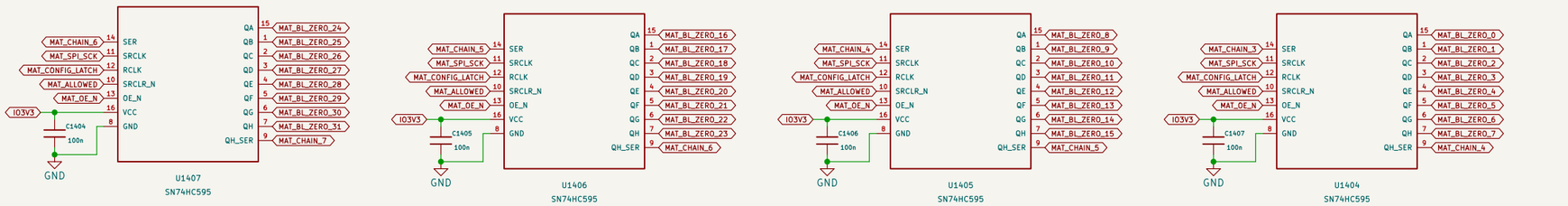
MIRA
Sheet: /MATRIX/1300_MATRIX_FPGA_CONTROL_AND_INTERLOCKS/
File: 1300_MATRIX_FPGA_CONTROL_AND_INTERLOCKS.kicad_sch
Title: Matrix commands and hardware interlocks
Size: A2 | Date: 2026-10-18 | Rev: B
KiCad E.D.A. 10.0.5 | Id: 19/30

Eighty-bit switch-control image

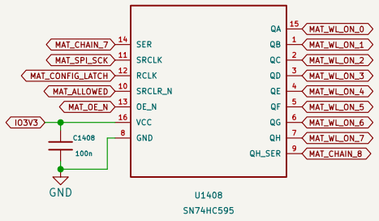
VMM MASK / 32 BL DAC SWITCHES



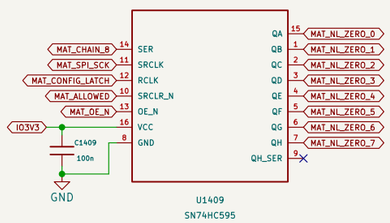
BL GROUND-CLAMP MASK / 32 CHANNELS



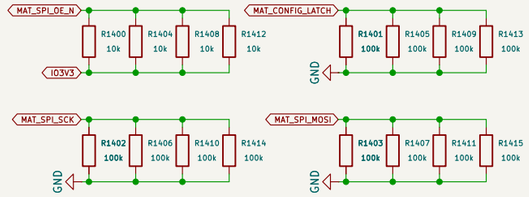
WL MASK



NL CLAMP MASK



COMMON SERIAL CONTROL / FOUR ORIGINAL PULL SETS



Serial order: U1400 -> U1401 -> U1402 -> U1403 -> U1404 -> U1405 -> U1406 -> U1407 -> U1408 -> U1409.
SCK shifts the serial register; LATCH copies all 80 bits to output storage. OE high makes outputs high-Z.

Serial chain and stored image

Ten SN74HC595 registers form one 80-bit chain, from U1400 through U1409. MAT_SPI_MOSI enters U1400, and MAT_CHAIN_0-8 connect successive serial outputs to the next register. MAT_SPI_SCK shifts the chain; MAT_CONFIG_LATCH transfers the complete image into the output storage registers.

Bit-line masks

U1400-U1403 supply the 32 MAT_BL_VMM_0-31 controls that connect individual DAC outputs to their bit lines. U1404-U1407 supply MAT_BL_ZERO_0-31 for the independent ground clamps. Firmware must avoid enabling a DAC connection and its ground clamp at the same time.

Word-line and neuron-line masks

U1408 provides MAT_WL_ON_0-7 for the eight word-line selectors. U1409 provides MAT_NL_ZERO_0-7 for the eight neuron-line ground clamps. The selected measurement line must have its clamp released before routing it to the analog return.

Two output-enable domains

The four VMM-mask registers use MAT_VMM_OE_N. The remaining six registers use MAT_OE_N. A high enable-control level makes the affected outputs high impedance. Downstream pull resistors then establish the individual switch-control defaults.

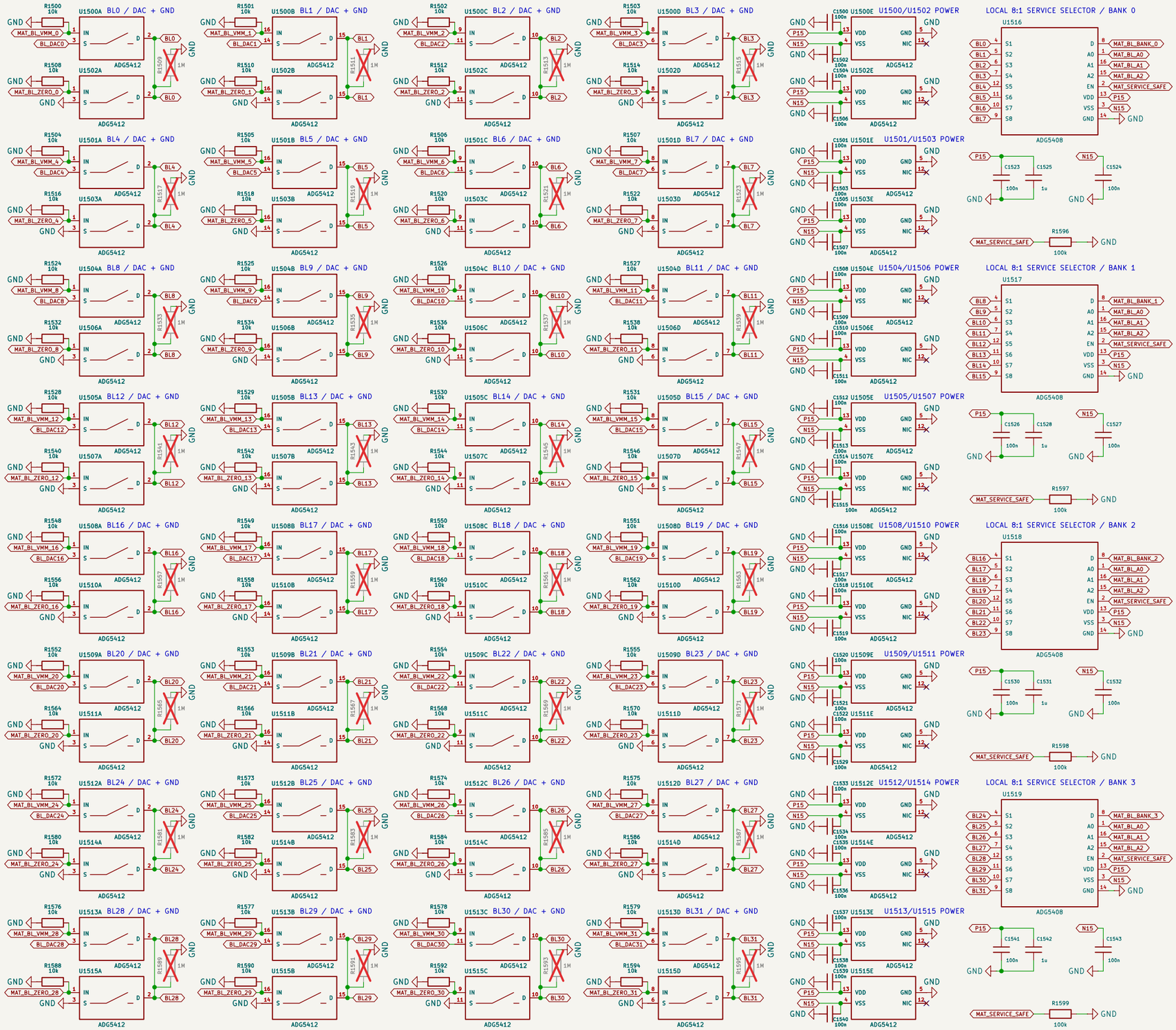
Clear and reload sequence

MAT_ALLOWED drives every active-low shift-register clear input. Clearing the shift register does not itself transfer cleared data into output storage. Keep outputs disabled, shift the full intended image, latch it, and only then request operation. The circuit includes local 100n bypass capacitors for all ten packages.

MIRA	
Sheet: /MATRIX/1400_MATRIX_SERIAL_CONTROL_REGISTERS/	
File: 1400_MATRIX_SERIAL_CONTROL_REGISTERS.kicad_sch	
Title: Eighty-bit switch-control image	
Size: A3	Date: 2026-10-18
KiCad E.D.A. 10.0.5	Id: 20/30

Rev: B

Bit-line DAC connections, clamps and service selection



Thirty-two bit-line terminals

This sheet contains BL0–BL31 in four groups of eight. Each terminal has two ADG5412 switch paths: one towards a buffered DAC source and one towards GND. The MAT_BL_VMM mask controls the DAC paths, while MAT_BL_ZERO controls the clamps. These two masks are generated by the serial image registers.

Normal drive and grounding

For drive, enable the DAC path and release its ground clamp; for GND, do the opposite. All 32 optional 1M BL bleeders are DNP to avoid unmeasured WRITE return current. With both switches OFF, a BL may float. Use BL_ZERO for a defined level or deliberate discharge. The 10k resistors bias control pins only.

Eight-to-one local selectors

U1516–U1519 each select one bit line from their eight-channel group. MAT_BL_A0, MAT_BL_A1 and MAT_BL_A2 are shared address inputs. The selector outputs are MAT_BL_BANK_0–3; the upper address selection on the service-router sheet chooses which bank reaches MAT_BL_SERVICE.

Service mode

MAT_SERVICE_SAFE enables the local service selectors. The service connection is separate from the VMM DAC switches, so firmware must prepare the bit-line masks before making a read, set, reset or forming connection. The chosen line must not be clamped against the active source.

Analog supply bypassing

The switch packages use P15 and N15 with local supply bypassing. BL6 and BL7 now use their dedicated BL_DAC6 and BL_DAC7 sources. All other DAC, clamp and service-selector connections are unchanged.

MIRA

Sheet: /MATRIX/1500_MATRIX_BL0_31_SWITCHING/

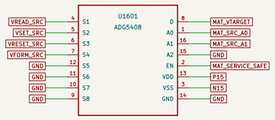
File: 1500_MATRIX_BL0_31_SWITCHING.kicad_sch

Title: Bit-line DAC connections, clamps and service selection

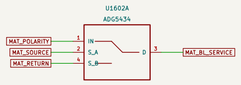
Size: A3	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.5		Id: 21/30

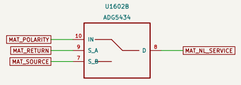
1600 / SERVICE COMMAND, ROUTING AND CONTROLLER RESET

VREAD / VSET / VRESET / VFORM COMMAND
SRC=00 READ, 01 SET, 10 RESET, 11 FORM.
U1601 no longer supplies DUT power.



DUT POLARITY
1: BL=SOURCE, NL=RETURN
0: BL=RETURN, NL=SOURCE





UPSTREAM POWER ENABLE
OFF: shunt input to GND, ON: CV/CC driver.
No pulse switch after the current shunt.

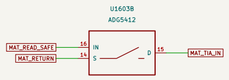


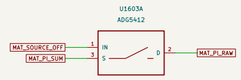


ERROR SELECTION
OFF selects zero; ON selects regulator error.
The PI controller and shunts are on sheet 2100.



RETURN PATH
READ: TIA virtual ground.
WRITE: direct GND; old low-side sink removed.

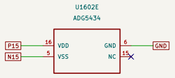




BL BANK SELECTION
Only the held service address is used.
VMM isolates the service path.

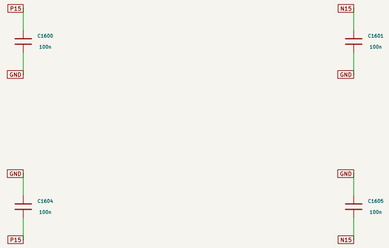


OFF RESET (MAT_SOURCE_OFF = NOT MAT_SOURCE_SAFE)
A clears the PI output-to-sum path.
D discharges the feedback capacitor directly.
R1605 limits direct capacitor-reset current.

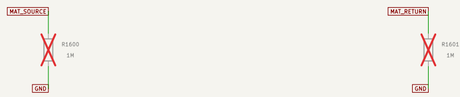




SUPPLIES / LOCAL BYPASS
All ADG54xx use P15/N15. Respect powered-rail sequencing.



DNP BLEEDERS / PASSIVE CONTROL PULL-DOWNS
R1600/R1601 must remain DNP for microamp operation. Power OFF discharges SOURCE through the selected shunt.
Before range, polarity or route changes: source OFF, discharge and wait: a loaded/open high-R range takes longer.

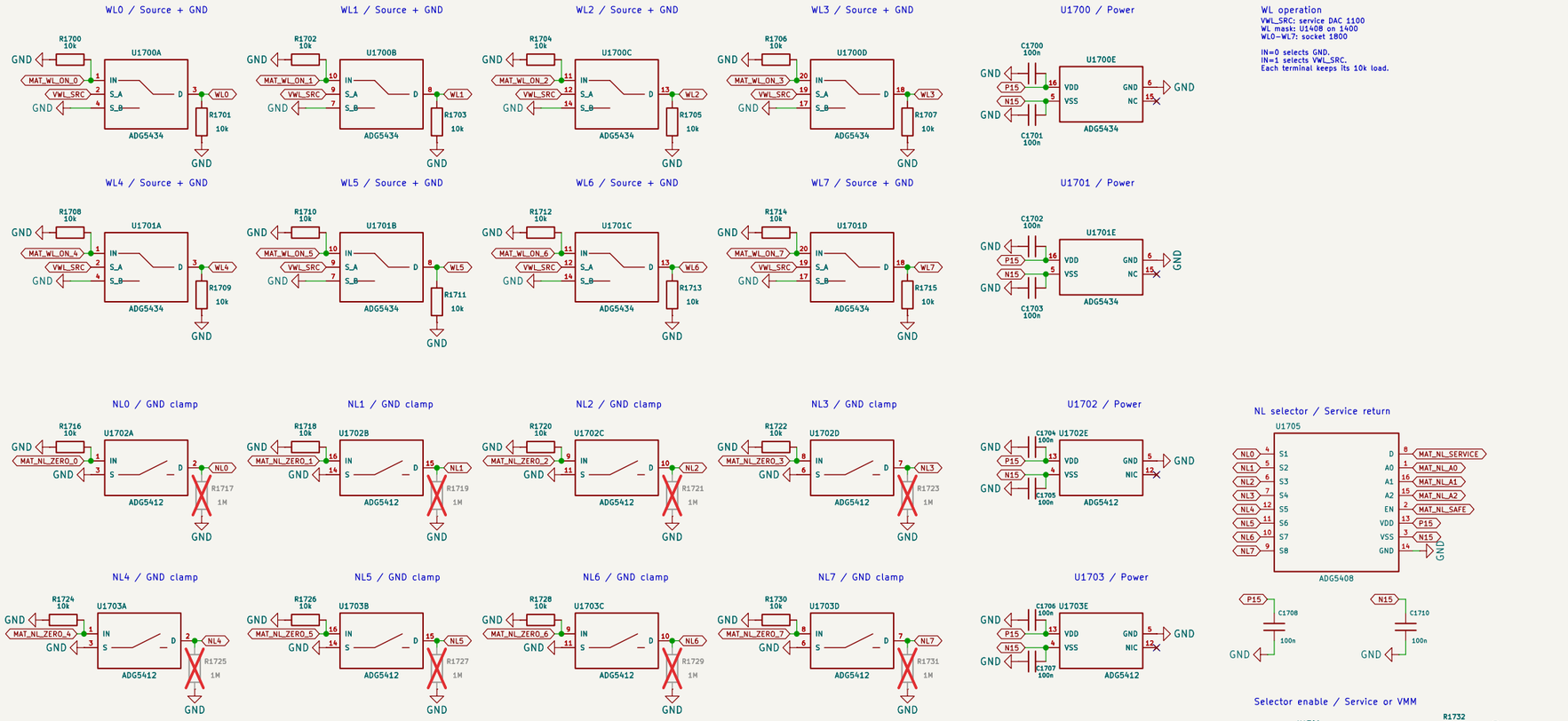


MIRA
Sheet: /MATRIX/1600_MATRIX_SERVICE_SOURCE_BANK_AND_POLARITY_ROUTING/
File: 1600_MATRIX_SERVICE_SOURCE_BANK_AND_POLARITY_ROUTING.kicad_sch

Title: Service source, bit-line bank and polarity routing

Size: A2	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 22/30

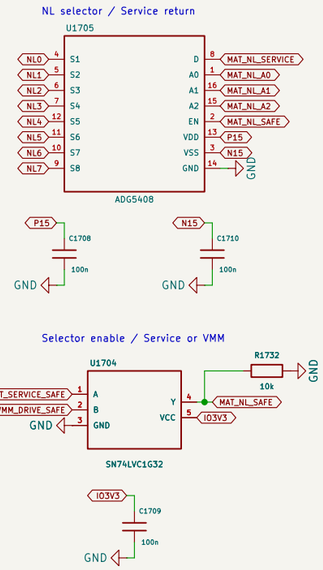
Word-line drive and neuron-line selection



WL operation
VWL_SRC: service DAC 1100
WL mask: U1408 on 1400
WL0–WL7: socket 1800

IN=0 selects GND.
IN=1 selects VWL_SRC.
Each terminal keeps its 10k load.

NL mask: U1409 on 1400
MAT_NL_SERVICE: polarity router 1600
One selected NL feeds the common TIA.
For VMM read, set MAT_POLARITY=1.
The selected NL clamp must be OFF.



Eight word-line selectors

U1700 and U1701 are quad ADG5434 switches. Their common outputs connect to WL0–WL7. Each channel selects between VWL_SRC and GND under the matching MAT_WL_ON_0–7 control from the serial register image. This supplies the selected word-line voltage to the crossbar transistor gates.

Word-line loading

Each WL terminal retains its 10k load to GND. The control inputs also have 10k bias resistors. VWL_SRC comes from the service DAC sheet, so changing that DAC command changes the common available word-line drive level rather than selecting an individual row.

Independent neuron-line clamps

U1702/U1703 provide one ground-clamp switch per NL, controlled by MAT_NL_ZERO_0–7. All eight optional 1M NL bleeders are DNP: a fitted resistor would bypass WRITE current sensing. Release the selected clamp for measurement; use NL_ZERO deliberately for a defined level or discharge.

Shared measurement selector

U1705 is an eight-to-one selector. MAT_NL_A0–2 choose one neuron line, and its common output is MAT_NL_SERVICE. U1704 generates MAT_NL_SAFE by ORing MAT_SERVICE_SAFE with MAT_VMM_DRIVE_SAFE; either qualified operating path can therefore enable the selector.

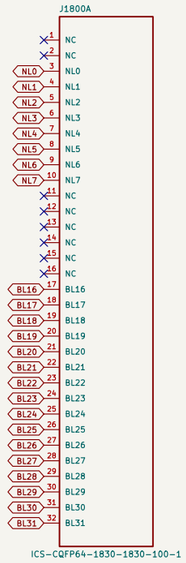
VMM return route

The selected neuron line continues through the polarity router and then to the common TIA during a read. The existing operating note specifies MAT_POLARITY=1 for VMM read. Only one NL is selected by this multiplexer at a time; the other neuron-line clamps remain independently programmable.

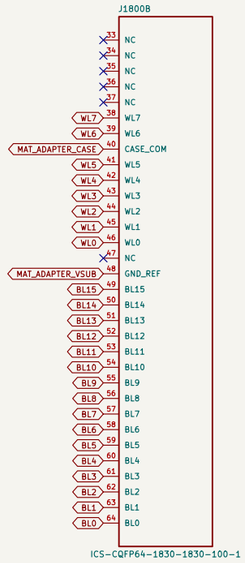
MIRA	
Sheet: /MATRIX/1700_MATRIX_WL0_7_DRIVERS_AND_NL0_7_CLAMPS_SELECTOR/	
File: 1700_MATRIX_WL0_7_DRIVERS_AND_NL0_7_CLAMPS_SELECTOR.kicad_sch	
Title: Word-line drive and neuron-line selection	
Size: A3	Date: 2026-10-18
KiCad E.D.A. 10.0.5	**Rev: B**
	Id: 23/30

Crossbar socket and bias

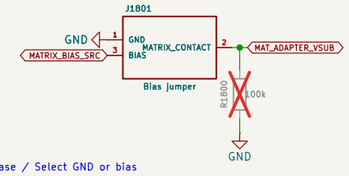
J1800 / Contacts 1–32



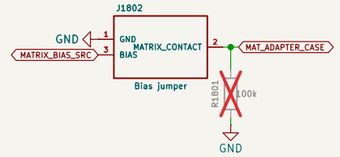
J1800 / Contacts 33–64



Pin 48 / Reference GND or bias



Case / Select GND or bias



One socket, two schematic units

J1800A and J1800B represent one ICS-CQFP64-1830-1830-100-1 socket. The two units divide contacts 1–32 and 33–64 for readability. Their pin numbers identify the same physical contacts in the assigned footprint; they are not two separate connectors.

Matrix signal assignment

BL0–BL15 use contacts 64–49 in descending order, while BL16–BL31 use contacts 17–32 in ascending order. NL0–NL7 use contacts 3–10. WL0–WL5 use contacts 46–41; WL6 and WL7 use contacts 39 and 38. These named nets connect to the BL switching and WL/NL driver sheets.

Case and reference selectors

Contact 40 is CASE_COM on MAT_ADAPTER_CASE. Contact 48 is GND_REF on the existing configurable MAT_ADAPTER_VSUB net. J1802 selects the case connection, and J1801 selects the contact–48 reference. For either jumper, pins 1–2 connect the matrix contact to GND; pins 2–3 connect it to MATRIX_BIAS_SRC. R1800/R1801 are optional 100 kOhm returns to GND; both are marked DNP and provide a return only when fitted.

Default and unused contacts

Use the ground positions for both bias selectors as the default. Apply external bias only when it matches the installed crossbar requirements. Contact 47 and the other NC contacts remain intentionally unconnected. The contact assignment follows the user crossbar reference; an unrelated source-board assignment must not override this mapping.

MIRA

Sheet: /MATRIX/1800_MATRIX_CROSSBAR_SOCKET_AND_BIAS/
File: 1800_MATRIX_CROSSBAR_SOCKET_AND_BIAS.kicad_sch

Title: Crossbar socket and bias

Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 24/30

ADC — Current conversion and precision acquisition

Consecutive circuit order; select a child sheet to inspect its circuit and sheet-specific explanation.

1900_ADC_TIA_AND_MONITOR_SELECTOR

TIA and monitor mux; high-impedance current feedback and ILIM copies; ILIM buffer also feeds shared source controller.

2000_ADC_CONVERTER_AND_PRECISION_REFERENCES

12 V preregulator, 5 V reference, balanced THP210 ADC input and AD4003.

Group scope

TIA ranges and monitor selection, high-impedance compliance buffers, balanced differential ADC drive, reference and converter.

Reading guide

Each numeric child contains its own circuit explanation. Identical global labels connect between leaves. A shared U reference denotes one physical multi-unit device.

Physical references follow the first circuit leaf containing the component; multi-unit devices are not duplicated.

MIRA

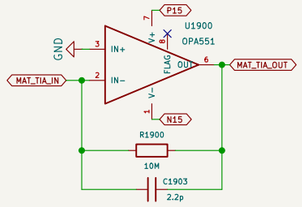
Sheet: /ADC/
File: ADC.kicad_sch

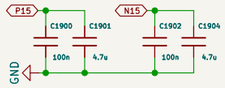
Title: Current conversion and precision acquisition

Size: A3	Date:	Rev: B
KiCad E.D.A. 10.0.5		Id: 25/30

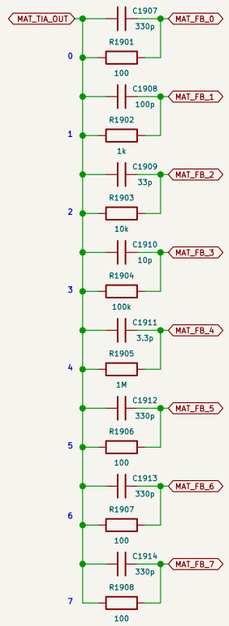
Matrix current conversion and monitor selection

TIA and permanent feedback

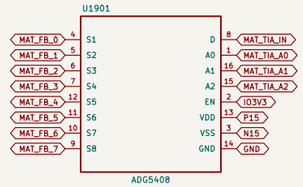


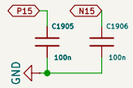


Paired Rf / Cf branches

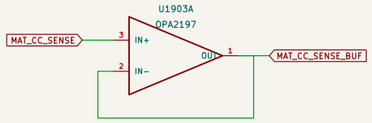


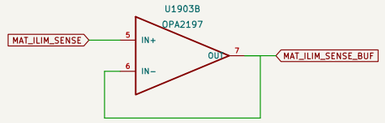
Feedback selector



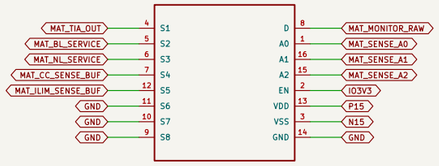


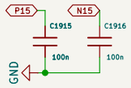
High−impedance taps for protection and monitor

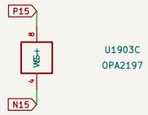




Voltage monitor selector







TIA gain and monitor address are independent



U1903: P15 / N15. Channel B also feeds the source regulator.

MAT_TIA_IN from service routing
MAT_MONITOR_RAW to ADC driver

Current−to−voltage conversion

U1900 holds MAT_TIA_IN near GND and converts input current into MAT_TIA_OUT. In the ideal low−frequency model VOUT = −IIN x REFF. REFF includes R1900 (10 MOhm) in parallel with the selected branch, including switch resistance. C1903 (2.2 pF) remains in feedback at every gain.

Selectable feedback branches

U1901 connects one paired resistor/capacitor branch across the amplifier. Address codes 0−4 select 100 Ohm/330 pF, 1 kOhm/100 pF, 10 kOhm/33 pF, 100 kOhm/10 pF, or 1 MOhm/3.3 pF. Codes 5−7 repeat the 100 Ohm/330 pF branch. Each selected branch operates in parallel with the permanent feedback network; switch resistance also affects the effective transimpedance.

Independent monitor address

U1902 selects TIA, BL service, NL service, buffered current sense or buffered ILIM. Codes 5−7 select GND. U1903A isolates the INA821 output; U1903B buffers the target for the controller, trip divider and monitor mux. The sheet 2000 buffer isolates the ADC gain network.

Measurement sequence and supplies

MAT_MONITOR_RAW feeds the high−input−impedance OPA197 follower on sheet 2000. Set the monitor and TIA range before conversion. Initial validation uses <=100 kSPS and a 20 us post−change delay, subject to analog settling verification. Analog buffers and switches use P15/N15.

MIRA

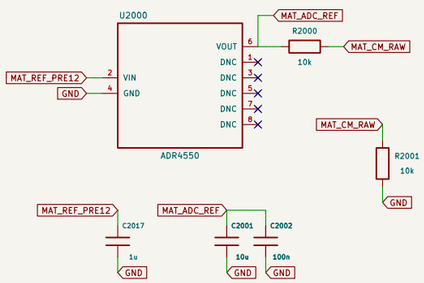
Sheet: /ADC/1900_ADC_TIA_AND_MONITOR_SELECTOR/
File: 1900_ADC_TIA_AND_MONITOR_SELECTOR.kicad_sch

Title: Matrix current conversion and monitor selection

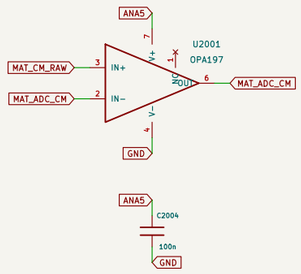
Size: A3	Date: 2026−10−18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 26/30

Buffered monitor and differential matrix ADC

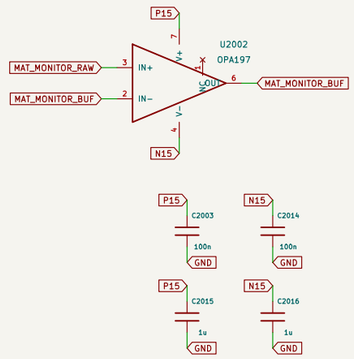
5 V reference from a protected 12 V input



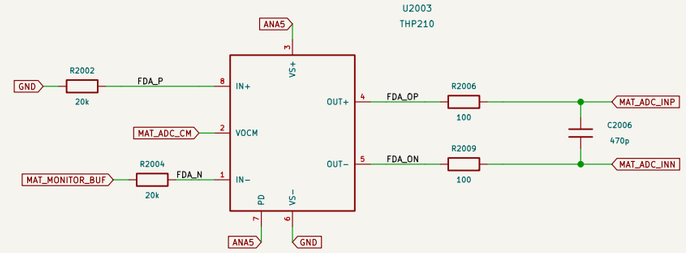
VREF / 2 common-mode buffer

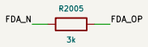


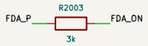
High-Z post-mux buffer: P15 / N15



Balanced FDA gain and ADC filter





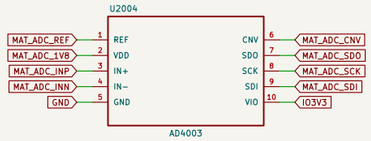


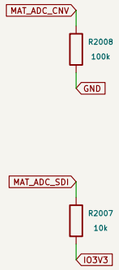
RF / RG = 3k / 20k = 0.15
Both ADC inputs move in antiphase:
C2006 is between IN+ and IN−.

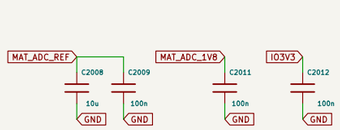


Matched 0.1% FDA resistor ratios.
PD is tied to ANA5 for normal operation.

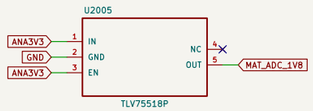
ADC and serial interface



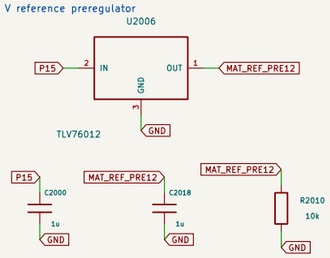




Local ADC 1.8 V supply



12 V reference preregulator



TLV76012 pin 1 OUT / pin 2 IN / pin 3 GND
R2010 guarantees a load above 1 mA.

Reference input headroom

U2006 derives a nominal 12 V input from P15 for U2000. This keeps the ADR4550 input away from its 15 V upper operating boundary. R2010 provides the preregulator minimum load; the 5 V reference is not powered from ANA5.

High-impedance monitor input

U2002 follows MAT_MONITOR_RAW on P15/N15 before the resistive FDA network. Its high-impedance input isolates the selected node from the FDA gain network. Raw CC and ILIM are separately buffered by U1903 on sheet 1900; S4/S5 observe those buffered nodes.

True differential conversion

U2003 is a THP210 powered by ANA5/GND. U2001 supplies VOCM = VREF/2. Ideal outputs are 2.5 V +/- 0.075 x VMON, giving VDIFF = -0.15 x VMON and a constant 2.5 V average. At VMON = +/-15 V the FDA outputs are 1.375...3.625 V and its input common mode is 1.196...3.152 V. The post-mux buffer requires rail headroom.

Filter and validation timing

R2006/R2009 are symmetric 100 Ohm isolation resistors; C2006 is 470 pF C0G between ADC inputs. Differential RC tau is 94 ns. Begin validation at <=100 kSPS with a 20 us delay after source/range changes. Confirm actual settling and loop stability; these initial settings are not a verified 2 MSPS specification.

Serial and local power

AD4003 uses MAT_ADC_1V8 for VDD and IO3V3 for its interface. CNV, SCK, SDI and SDO retain their U500 connections. R2008 holds CNV low; R2007 holds SDI high. Use the gain -0.15 in conversion and per-range calibration.

MIRA

Sheet: /ADC/2000_ADC_CONVERTER_AND_PRECISION_REFERENCES/
File: 2000_ADC_CONVERTER_AND_PRECISION_REFERENCES.kicad_sch

Title: Buffered monitor and differential matrix ADC

Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 27/30

PROTECTION — Shared CV/CC source and hardware fault memory

Consecutive circuit order; select a child sheet to inspect its circuit and sheet-specific explanation.

2100_PROTECTION_WRITE_CURRENT_COMPLIANCE

Source-side CV/CC PI, four Kelvin shunt ranges, current/voltage errors, command clamps and back-calculation anti-windup.

2200_PROTECTION_MATRIX_TRIP_AND_FAULT_INTERFACE

Backup overcurrent threshold with offset margin; configuration-trip OR, fault latch, qualified clear and external inhibit.

Group scope

Source-side Kelvin current sensing, shared PI control with bounded command and reset, independent backup overcurrent trip, configuration fault latch and external inhibit.

Reading guide

Each numeric child contains its own circuit explanation. Identical global labels connect between leaves. A shared U reference denotes one physical multi-unit device.

Model limits: downstream stored charge can still discharge through the DUT. Use the documented current, capacitance, settling and reconfiguration contract; qualify final PCB transients on the bench.

Physical references follow the first circuit leaf containing the component; multi-unit devices are not duplicated.

MIRA

Sheet: /PROTECTION/
File: PROTECTION.kicad_sch

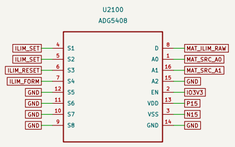
Title: Shared CV/CC source and hardware fault memory

Size: A3	Date:	**Rev: B**
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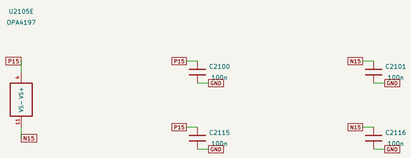
Bias = ANA5/2; command ceiling = 2.38 x MAT_ADC_REF (nominal 11.9 V). All analog amplifiers use P15 / N15.

Scope: source-side CV/CC. Output current includes DUT, leakage and line charging. No intentional reservoir after RSH.

Current command selection



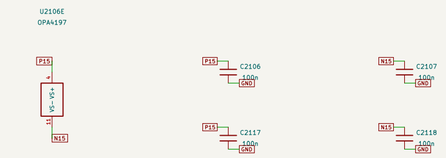
00 READ / 01 SET / 10 RESET / 11 FORM
READ and SET share ILIM_SET. Reload only with outputs OFF.



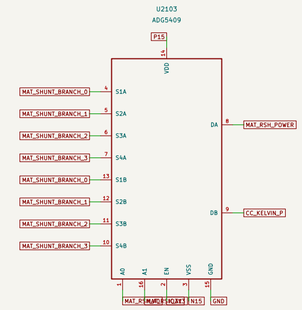
Four source shunts



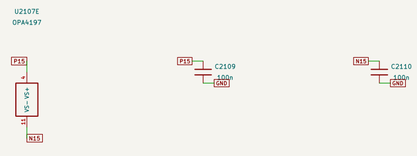
RSH: 100 / 1k / 10k / 100k
At 0.1.1 V: 1..10 mA / 0.1..1 mA /
10..100 uA / 1..10 uA. Calibrate each range.



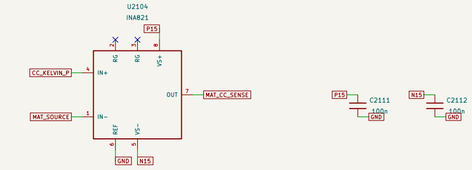
Kelvin range selector

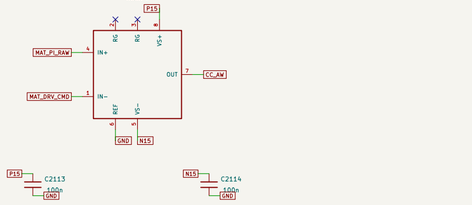


A = force path; B = high-Z Kelvin sense
EN stays high; switch ranges only with source OFF.



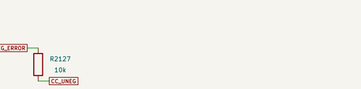
Current and back-calculation sense





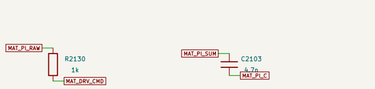
Bias reference

BIAS is shared by both error cells and cancelled before the PI.



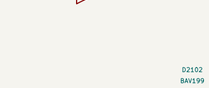
Common-mode ceiling

11.9 V ceiling uses the ADC precision reference through a high-Z input.



Command sense buffer

CBUF isolates CMD from clamp feedback resistor loads.





Voltage sense and source driver

Voltage input is high-Z. Driver feeds source-OFF SPDT on sheet1600.

Voltage error

VERR = BIAS + VSOURCE − VTARGET.

Current error

IERR = BIAS + ISENSE − buffered ILIM target.

Precision maximum: voltage

D2100 provides main and local anti-saturation feedback.

Precision maximum: current

D2101 selects the larger shifted error without a floating branch.

Remove error bias

ERROR = max(VERR, IERR) − BIAS.

Shared PI and back-calculation

Sheet1600 resets PI_SUM to RAW and to C in source-OFF state.

Command lower clamp

CBUF feedback keeps the clamp branch linear while inactive.

Lower limit 0 V; upper command nominal 11.9 V. CBUF isolates clamp feedback. INA821 back-calculation measures RAW − CMD without loading CMD.

Command upper clamp

Ceiling = 2.38 x MAT_ADC_REF; actual output depends on load drop.

MIRA

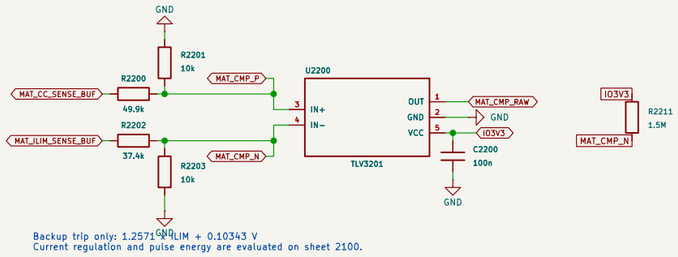
Sheet: /PROTECTION/2100_PROTECTION_WRITE_CURRENT_COMPLIANCE/
File: 2100_PROTECTION_WRITE_CURRENT_COMPLIANCE.kicad_sch

Title: Source-side CV/CC, Kelvin current sense and anti-windup

Size: A2	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.5		Id: 29/30

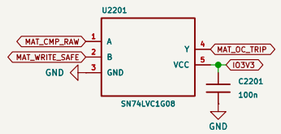
Write overcurrent trip, fault memory and power inhibit

Sense dividers and comparator

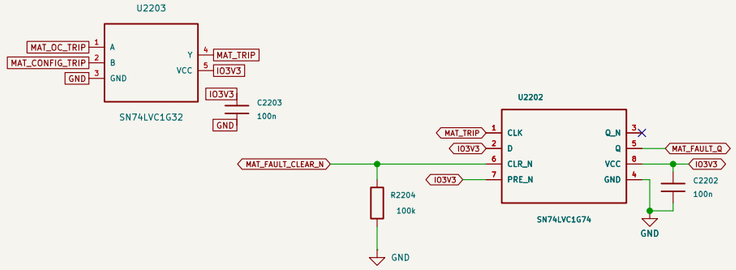


Backup trip only: 1.2571 * ILIM + 0.10343 V
Current regulation and pulse energy are evaluated on sheet 2100.

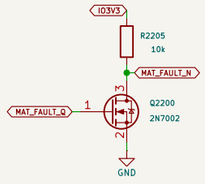
Qualified trip event



Clocked trip memory / asynchronous clear

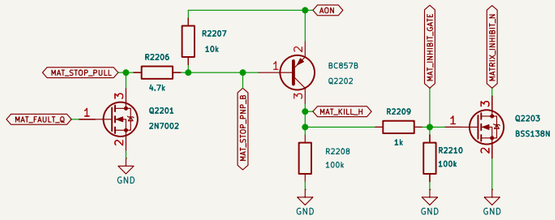


FPGA fault output



MAT_FAULT_N to FPGA Interlock

Power−module inhibit driver



MATRIX_INHIBIT_N to Power / connector sheet 400

Protected overcurrent scale

R2200/R2201 = 49.9k/10k divide normalized source−current sense by 5.99. R2202/R2203 = 37.4k/10k and R2211 = 1.5M from IO3V3 set the backup threshold. Nominal trip: VSENSE > 1.2571*VILIM + 0.10343 V. The CV/CC loop is the precision current regulator; this comparator is independent coarse shutdown.

Input headroom and fault current

At +16.5 V sense, IN+ is at most about 2.76 V, inside the valid common−mode range for IO3V3 >= 3.135 V. Negative saturation or a powered−off comparator is current limited by the input resistors; calculated clamp current is below 0.5 mA versus the 10 mA absolute limit. Normal source sense and limit are 0..1 V. Do not use the fault comparator as a calibrated low−current meter.

Analog and configuration trips

U2201 keeps analog qualification: MAT_OC_TRIP = MAT_CMP_RAW & MAT_WRITE_SAFE. U2203 ORs this event with MAT_CONFIG_TRIP from sheet 1300. Route/range/mode changes and service−DAC updates during intended WRITE now clock the same fault memory as excess current. The configuration branch is not masked by WRITE_SAFE.

Fault memory and re−arm

U2202 stores D=1 on a MAT_TRIP rising edge. PRE_N stays high; only CLR_N is asynchronous. R2204 defaults clear low, also disabling MAT_ALLOWED on sheet 1300. Clear release alone cannot arm the matrix; source zero, a full validated 80−bit frame and protected LATCH are required. Meet latch clock−pulse and clear recovery/removal timing.

Local stop and Power inhibit

MAT_FAULT_Q pulls MAT_FAULT_N low through Q2200 and drives Q2201/Q2202/Q2203 to assert MATRIX_INHIBIT_N to Power. Local enable removal does not wait for MCU or ADC. Total shutdown delay includes sense amplifier, input RC, comparator, gates, latch, source actuator and switch timing; the 40 ns comparator headline is not the system stopping time.

MIRA
Sheet: /PROTECTION/2200_PROTECTION_MATRIX_TRIP_AND_FAULT_INTERFACE/
File: 2200_PROTECTION_MATRIX_TRIP_AND_FAULT_INTERFACE.kicad_sch
Title: Write overcurrent trip, fault memory and power inhibit
Size: A3 | Date: 2026−10−18 | **Rev: B**
KiCad E.D.A. 10.0.5 | Id: 30/30